

PT GALAKSI MEGATAMA INDONESIA

Manufacturer of record

TECHNICAL CERTIFICATION DOCUMENT

IECEx/ATEX Certification Document

Gas Leak Detector (GLD) V2

Prepared in direct reference to the IECEx/ATEX Certification Information Requirements issued by the certification body (ExCB) — covering all three sections of that checklist: Section 1, Basic Information (Application and Organization); Section 2, Technical Documentation (Items 1–6: product description; name, model, and specification list; functional description and technical parameters; product photographs; intended use and installation environment; and design and manufacturing information); and Section 3, Sample Information.

Document Control

Document title	IECEx/ATEX Certification Document — Gas Leak Detector (GLD) V2
Document no.	GMI/GLD/IECEX-TDF/2026-001
Revision	1.5
Date	7 October 2026
Status	Working Document — Draft for Internal Review
Classification	Confidential — prepared for ATEX/IECEx certification body (ExCB) submission
Certification subject	Node Sensor (GLD) — V2
Applicant / Manufacturer	PT Galaksi Megatama Indonesia
Reference checklist	IECEx/ATEX Certification Information Requirements

Revision History

REVISION	DATE	DESCRIPTION
0.1	11 September 2026	Initial issue — Section 2 (Items 1–6) and Section 3.
0.2	17 September 2026	Section 1, Basic Information (Application and Organization), added — Items 1.1–1.5. Document now covers all three sections of the checklist.
0.3	22 September 2026	Manufacturer-of-record for Section 1 changed to PT Galaksi Megatama Indonesia (legal documents on file). Section 2.6.g draft content (safety

		warnings, installation procedures, operating modes, commissioning equipment) added.
0.4	22 September 2026	Candidate enclosure component data (ADC12 die-cast alloy, M20×1.5 cable entry, electrostatic-safety parameters) added to Sections 2.3.c, 2.6.a, and 2.6.c, sourced from a supplier reference drawing in the casing partner's supply chain.
0.5	22 September 2026	Enclosure material grade confirmed as ADC12 (Section 2.3.c, status raised to Final with an explicit caveat that this extrapolates from the casing manufacturer's sub-component drawing, not a direct callout on the main enclosure drawing). Cable entry (Section 2.3.c) confirmed as a 2-conductor cable labeled L+/L-, each conductor ≈0.75 mm diameter.
0.6	22 September 2026	Total weight corrected to 2.3 kg; IP rating confirmed as IP66; operating humidity confirmed as 10–90% RH non-condensing (standard industrial rating, matched to a comparable certified product); cable gland confirmed as M20×1.5 (IP66-rated) — all raised to Final (Section 2.3.c). Removed the ambiguous “no plastic or PVC” parenthetical from the product description (Section 2.1) and generalized “L-bracket” to “bracket” in that same narrative sentence (the dimensioned bracket/U-bolt specification in Section 2.3.c is unaffected). “GLD V2 (Version 2)” simplified to “GLD V2”.
0.7	22 September 2026	“L-bracket” terminology corrected throughout to “U-bolt mounting plate” (Sections 2.1, 2.3.c, 2.6.g), matching the actual CAD drawing (a flat 250×250 mm plate secured by 2× U-bolts — not an angled/L-shaped bracket).
0.8	24 September 2026	Total weight, operating temperature, and operating humidity (Sections 2.3.c, 2.5) updated to 2.378 kg, –20°C to

		+60°C, and 5–95% RH respectively, per current project specification data.
0.9	25 September 2026	Section 2.6.c (material specification sheets, non-metallic materials) expanded with the specific IEC 60079-0 requirements (CTI class, glow-wire/flame-retardancy, Tg vs. the confirmed operating range, UV/weathering resistance) and a list of the non-metallic components in GLD's own design that this item will need to cover (sensor mesh/flame-arrestor cover, cover-to-body sealing gasket, cable-gland seal, antenna feedthrough insulator); status remains Not yet available pending the casing partner's data.
1.0	25 September 2026	Section 2.6.c updated with materials confirmed from a newer casing-partner assembly drawing: the protective mesh cover is stainless steel (metallic, outside the scope of this non-metallic-materials item) and the cover-to-body seal is a rubber gasket (compound/grade still not specified).
1.1	25 September 2026	Added an explicit equipment-boundary statement to Section 2.3.a: the Node Sensor (GLD) receives only 24 VDC at its terminal and never carries 220 VAC internally; the AC/DC adapter is a separate, site-supplied device outside the certified unit's physical boundary. Section 2.6.c cable-gland seal remark clarified as specified together with the M20×1.5 gland itself from the BP18-1Z reference, per project direction, rather than as a separately sourced item.
1.2	28 September 2026	Type of protection narrowed to a single team recommendation, Ex d (flameproof enclosure), in Sections 1.1 and 2.5 — selected over Ex e and Ex i based on the existing ADC12 die-cast enclosure with a stainless-steel mesh flame path (matching the three certified reference products reviewed) and the MQ-sensor heaters' continuous, unregulated power draw

		being difficult to reconcile with intrinsically safe energy limits for Group IIC. Section 2.6.e updated accordingly. Still an internal recommendation, not an ExCB decision.
1.3	28 September 2026	Applicant/Manufacturer identity simplified to a single entity, PT Galaksi Megatama Indonesia, throughout the document (document number prefix changed to GMI/...). The separate “design and development authority” and “technical partner” entries were removed (Sections 1.1, 1.3, 2.2, letterhead, Document Control); narrative passages describing electronics design/engineering work were made generic rather than deleted, to avoid implying the Manufacturer performed work it did not.
1.4	30 September 2026	Section 2.6.f populated with a preliminary internal surface-temperature measurement (eight thermocouple channels; normal, maximum-load and fan-stalled conditions; 180 minutes each), including extrapolation to the +60°C maximum ambient: highest value ≈114°C (MQ sensor body, fan stalled) against an effective T4 limit of 130°C. Status raised from Not yet available to Partially available; open measurement items listed. Sections 1.1 and 2.5 temperature-class remarks updated accordingly.
1.5	7 October 2026	Section 2.6.a extended with the released main-board electrical schematic (26 sheets, Section 2.6.a.7) and ten annotated PCB layout views with board data (Section 2.6.a.8), including the protection-device list, grounding description and open verification items. Schematic status raised to Available; PCB layout remains Partially available.

Table of Contents

About This Document.....	7
1. Basic Information (Application and Organization).....	7
1.0 · Status Summary — Section 1.....	8
1.1 · Application Form.....	8
1.2 · Business License / Company Registration Certificate.....	9
1.3 · Organizational Chart and Contact Information.....	10
1.4 · Manufacturing Plant Address and Introduction of Production Facilities.....	11
1.5 · ISO 9001 Certificate, Quality Manual, and Procedure Index.....	12
2. Technical Documentation.....	12
2.1 · Detailed Product Description.....	12
2.2 · Product Name, Model, and Specification List.....	13
2.3 · Functional Description and Technical Parameters (Electrical, Mechanical, etc.).....	13
2.3.a · Electrical Parameters — Node Sensor (GLD).....	14
2.3.b · Sensor and Communication Parameters — Node Sensor (GLD).....	15
2.3.c · Mechanical Parameters — Node Sensor (GLD).....	16
2.4 · Product Photographs — Overall and Key Components.....	19
2.5 · Description of Intended Use and Installation Environment (Gas Group, Temperature Class, Area Classification).....	20
2.6 · Design and Manufacturing Information.....	22
2.6.a · Complete Drawings (Assembly, Component, Electrical Schematic, PCB Layout, Enclosure Structure, Junction Box, Terminal, Grounding).....	22
2.6.b · Bill of Materials (BOM) for Explosion-Safety-Relevant Components.....	69
2.6.c · Material Specification Sheets / Datasheets (Non-Metallic Materials).....	72
2.6.d · Manufacturing Process Description.....	73
2.6.e · Explosion-Protection Calculations and Explanations (if applicable).....	73
2.6.f · Temperature Group Calculation (Hottest-Point Temperature Estimation).....	73
2.6.g · Usage and Installation Instructions (Draft).....	74
2.6.h · Nameplate Information.....	76
2.6.i · Ex Component Certificates.....	76
3. Sample Information.....	76
3.1 · Model, Serial Number, and Status.....	76
3.2 · Test Fixtures and Auxiliary Equipment.....	77

About This Document

This document is a working technical file prepared against the official IECEx/ATEX Certification Information Requirements checklist issued by the certification body (ExCB). It follows that checklist section by section: “1. Basic Information (Application and Organization)”, “2. Technical Documentation”, and “3. Sample Information”. Section 1 is administrative: it sets out every particular the ExCB requires and states which are already confirmed and which remain to be supplied from corporate records, without entering assumed values.

Scope of this document. This is a compilation of design evidence and an honest readiness assessment against the ExCB checklist, prepared in support of a future submission — it is not itself a certificate and does not constitute self-certification. The final enclosure/casing (material selection, manufacture, gasket, and cable entry) is being developed by an external mechanical/casing partner (PT Galaksi Megatama Indonesia — legal identity confirmed, Section 1.4) and is not yet in the authors' possession; items that depend on that design (material datasheets, manufacturing process description, explosion-protection calculations) are reported as not yet available for that reason, not because the work has stalled. The product is still at from-scratch development/prototype stage — no finalized, serialized units exist yet, which is why a formal sample register (Section 3.1) is not yet available either.

Original excerpt, Section 2 & 3 — source: IECEx ATEX Certification Information Requirements (ExCB):

“1) Detailed product description; 2) Product name, model, and specification list; 3) Complete and clear functional description and technical parameters (electrical parameters, mechanical parameters, etc.); 4) Clear product photos (overall and key components); 5) Description of intended use and installation environment (e.g., gas group IIC/IIB/IIA if applicable, temperature group T1–T6, ambient temperature range, area classification 0/1/2 or 20/21/22); 6) Design and manufacturing information [a–i: drawings, BOM, material datasheets, manufacturing process, explosion-protection calculations, temperature group calculation, usage and installation instructions, nameplate information, Ex component certificates]. 3. Sample Information: 1) model, serial number, and status of the sample; 2) necessary test fixtures or auxiliary equipment.”

1. Basic Information (Application and Organization)

Five items per the original checklist: the application form, the business license/company registration certificate, the organizational chart and contact information, the manufacturing plant address and production-facility profile, and — where applicable — the ISO 9001 certificate together with the quality manual and procedure index.

How this section is presented. The items below are administrative rather than technical: they are satisfied by corporate records held by the manufacturer's legal and administrative functions, not by engineering output. Every required field is therefore set out in full so that it can be collected in a single pass, with its current status stated plainly. No field has been filled with an assumed or placeholder value.

Applicant and Manufacturer of record for this document: PT Galaksi Megatama Indonesia. This section reports the legal/corporate particulars of PT Galaksi Megatama Indonesia, since it holds the Indonesian industrial manufacturing business license (NIB) relevant to fabricating the equipment (KBLI 25119/25120/28221/25920 — metal products, pressurized-vessel-class containers, metalworking machinery, and special metal-treatment services). PT Galaksi Megatama Indonesia is recorded as the sole applicant and manufacturer throughout this document. This attribution reflects a working decision recorded 28 September 2026 based on PT Galaksi's legal documents (deed of establishment, NIB, NPWP, MOLHR ratification decree, tax registration certificate) supplied to the authors — it has not yet been confirmed directly with the company's officers as the final arrangement for the ExCB submission and

should be verified before formal filing.

Original excerpt, Section 1 — source: IECEx ATEX Certification Information Requirements (ExCB):

“1. Basic Information (Application and Organization): 1) Application form (provided by ExCB as a template). 2) Manufacturer's business license/company registration certificate. 3) Manufacturer's organizational chart and contact information. 4) Address of manufacturing plant and brief introduction of production facilities. 5) (If applicable) ISO 9001 certificate, quality manual, and directory of procedure documents (for QAR/QAN review).”

1.0 • Status Summary — Section 1

ITEM	STATUS	HELD BY
1.1 Application form (ExCB template)	Template not yet received	Issued by the ExCB
1.2 Business license / company registration	Partially available	Manufacturer (PT Galaksi Megatama Indonesia) — legal documents now on file; a few particulars still open
1.3 Organizational chart and contact information	Partially available	Manufacturer — Board of Directors/Commissioners names known from the deed; a full org chart and named ExCB contacts remain to be provided
1.4 Manufacturing plant address and facility profile	Partially available	Manufacturer's registered office and production-site addresses now on file; floor profile, process capability, and quality system still pending
1.5 ISO 9001 certificate, quality manual, procedure index	To be confirmed	Manufacturer — quality function

1.1 • Application Form

The application form is issued by the certification body as a template and has not yet been received; it is therefore not reproduced here. The information needed to complete it, however, is already consolidated in this document, and is restated below in the order an application form normally requests it so that transcription is a single step once the template arrives.

APPLICATION FIELD	VALUE	STATUS	REFERENCE
Applicant / manufacturer	PT Galaksi Megatama Indonesia	Confirmed	Section 1.2
Product name	Gas Leak Detector (GLD) — Node Sensor	Confirmed	Section 2.2
Model / version	GLD V2	Confirmed	Section 2.2
Scope of certification	Node Sensor (GLD) only	Confirmed	Section 2.1

APPLICATION FIELD	VALUE	STATUS	REFERENCE
Requested area classification	Zone 1, Equipment Category 2G, Group II	Requested — subject to ExCB assessment	Section 2.5
Requested temperature class	T4 (≤ 135 °C)	Target — preliminary internal measurement supports it; formal verification outstanding	Sections 2.5, 2.6.f
Requested gas group	IIC	Internal recommendation, not an ExCB decision	Section 2.5
Type of protection	Ex d — flameproof enclosure	Internal recommendation, not an ExCB decision	Section 2.5
Standards to be applied	IEC 60079-0, together with IEC 60079-1 (flameproof enclosures “d”)	Dependent on the item above	Section 2.6.e
Certification route requested	IECEX Certificate of Conformity and/or ATEX EU-type examination	To be confirmed with the ExCB	
Intended markets	To be provided	To be provided	

The three requested classification parameters above are the applicant's proposal. They are recorded here as a request, not as an agreed or granted classification.

1.2 • Business License / Company Registration Certificate

REQUIRED PARTICULAR	STATUS	REMARKS
Registered legal name	Available	PT Galaksi Megatama Indonesia.
Legal form and shareholding status	Available	Perseroan Terbatas (limited liability company), National Private / PMDN (domestic investment), Small Enterprise scale. Deed No. 52 (22 Sep 2022, Notary Steffi Alphanie, S.H., M.Kn., Cilegon City): President Director Nur Rohman (2,250 shares), Director Antonius Prasetyo (2,625 shares), Commissioner Crisa Andi Sujatmiko (2,625 shares); authorized capital Rp300,000,000, paid-up Rp75,000,000.
Business registration number	Available	NIB (Business Identification Number) 0510220020593, issued 5 Oct 2022 under Indonesia's risk-based business licensing (OSS) system.
Taxpayer identification number	Available	NPWP 61.126.496.1-427.000 (16-digit: 0611264961427000), Tax Service Office of Pratama West Bekasi, registered 29 Sep 2022.

REQUIRED PARTICULAR	STATUS	REMARKS
Deed of establishment and latest amendment	Partially available	Deed of Establishment No. 52 dated 22 Sep 2022, ratified by Ministry of Law and Human Rights Decree No. AHU-0067398.AH.01.01.Year 2022 (29 Sep 2022, Company Register No. AHU-0194149.AH.01.11.Year 2022). No subsequent amendment deed has been supplied; whether one exists is unconfirmed.
Registered (domicile) address	Available	Plaza Summarecon Bekasi, Jl. Bulevar Ahmad Yani Kav. K.01, Level 7, Harapanmulya Village, Medansatria Sub-district, Bekasi City, West Java, Postal Code 17143.
Scope of business activity relevant to this product	Partially available	NIB lists KBLI 25119 (other prefabricated metal products for construction), 25120 (tanks, reservoirs and containers of metal – explicitly including pressurized gas cylinders), 28221 (machinery/machine tools for metalworking), and 25920 (special metal-treatment industrial services), among others, at production sites in Cikarang Selatan and Mustikajaya, Bekasi. This is a general metal-fabrication license; it does not by itself constitute an Ex-specific manufacturing qualification.
Scanned certificate, with English translation where requested	Available	NIB, NPWP, Deed of Establishment, MOLHR ratification decree, and Certificate of Registration (SKT) are on file with sworn English translations (Fikri Said Obed, S.S., Sworn Translator, Jakarta, 19 Sep 2026). Notarization/legalization requirements for ExCB submission to be confirmed.

1.3 • Organizational Chart and Contact Information

REQUIRED PARTICULAR	STATUS	REMARKS
Organizational chart	Partially available	Statutory Board of Directors/Commissioners known from the deed of establishment (below); a functional chart showing the units responsible for design, production, and quality, and the reporting line between them, has not been provided.
Board of Directors and Board of Commissioners	Available	President Director: Nur Rohman. Director: Antonius Prasetyo. Commissioner: Crisa Andi Sujatmiko. (Deed No. 52, 22 Sep 2022.)
Authorized signatory for the application	To be confirmed	The President Director (Nur Rohman) is the statutory officer entitled to represent the company (Articles of Association, Art. 12.5.a); whether he is the specific individual who will sign the ExCB application is not yet confirmed.
Certification project contact	To be provided	Name, position, e-mail, and telephone — the single point of contact for ExCB correspondence.

REQUIRED PARTICULAR	STATUS	REMARKS
Technical contact for the product	To be provided	The engineer who will answer technical queries on the technical file.
Quality contact	To be provided	Counterpart for the quality assessment referred to in 1.5.
Correspondence address and working language	To be provided	English is assumed for correspondence with the ExCB unless stated otherwise.

1.4 • Manufacturing Plant Address and Introduction of Production Facilities

The Manufacturer's registered office and NIB-listed production-site addresses are now on file (below). Facility floor profile, process capability, and quality-system detail are not yet documented. The product is at prototype stage, and the specific location where serial units will be produced has not been fixed — this is stated as a fact of the current development phase, not as an omission from the file.

PRODUCTION ELEMENT	STATUS	REMARKS
Manufacturer's registered office and production sites	Available	PT Galaksi Megatama Indonesia — registered office: Plaza Summarecon Bekasi, Jl. Bulevar Ahmad Yani Kav. K.01, Level 7, Bekasi City, West Java. NIB-registered production sites: Jl. Kp. Jatipilar, Cikarang Selatan, Bekasi Regency; and Jl. WR Supratman, Mustikajaya, Bekasi City. NIB lists metalworking KBLI including 25120 (Manufacture of Tanks, Reservoirs and Containers of Metal – explicitly covers pressurized gas cylinders), 25119, 28221 (metalworking machinery), and 25920 (special metal-treatment services). This confirms a licensed general metal-fabrication business — it is not an Ex-certified enclosure manufacturer license, and no ISO 9001 or Ex-specific quality certificate has been supplied.
Printed circuit board fabrication and assembly	Partially available	The electronic design is maintained as a native EasyEDA/JLCPCB project and the component supply chain is referenced to LCSC supplier part numbers throughout the bill of materials (Section 2.6.b). The fabrication and assembly provider, and its address, remain to be confirmed in writing.
Enclosure / casing manufacture	Partially available	Enclosure development and manufacture sits with the Manufacturer, PT Galaksi Megatama Indonesia (identity and registered addresses above). Plant floor profile, process capability, and Ex-specific process controls remain to be provided (see remaining rows below and Sections 2.6.c/2.6.d).
Final assembly, configuration, and functional test	Prototype stage	Currently carried out in the development laboratory environment. The production location for serial units has not been fixed.

PRODUCTION ELEMENT	STATUS	REMARKS
Production facility profile	To be provided	Floor area, principal equipment, staffing, and nominal throughput — the “brief introduction” requested by the checklist.
Incoming inspection and final quality control	To be provided	Arrangements for verifying purchased components and finished units.
Ex-specific routine verification	To be provided	Routine tests and verifications required of production units once a type of protection is selected.

Where any stage of serial manufacture is subcontracted, certification bodies normally expect each manufacturing location covered by the quality assessment to be identified by name and address. The final list of locations therefore depends on the production arrangement adopted after the prototype stage.

1.5 • ISO 9001 Certificate, Quality Manual, and Procedure Index

REQUIRED PARTICULAR	STATUS	REMARKS
ISO 9001 certificate	To be confirmed	Whether the manufacturer currently holds certification and, if so, the certificate number, scope, issuing body, and validity period.
Quality manual	To be provided	
Index of procedure documents	To be provided	A directory is sufficient at this stage; individual procedures are normally requested during the assessment itself.
Documented control of Ex-relevant characteristics	To be provided	How drawings, changes, and the characteristics that secure the type of protection are controlled in production.
Records of routine verification	To be provided	Record format for the routine tests referred to in 1.4.

Note on the quality requirement. The checklist marks this item “if applicable” and links it to quality assessment review. Under both schemes a manufacturer is expected to demonstrate a quality system covering Ex production before certificates are issued — through an IECEx Quality Assessment Report, or through the corresponding production-quality or product-verification arrangement under ATEX — whether or not ISO 9001 certification is held. The route that applies to this application is to be confirmed with the ExCB; ISO 9001 certification, where held, generally shortens that assessment rather than replacing it.

2. Technical Documentation

Six items per the original checklist (1–6, with Item 6 comprising sub-items a–i). This revision addresses the complete section: Items 1–6.

2.1 • Detailed Product Description

The Gas Leak Detector (GLD) is an IoT-based, multi-sensor gas leak detection device designed for the early detection of flammable and process gases in oil & gas refinery environments (process units, tank farms, pipe

racks, and storage/loading-unloading areas). This document addresses the GLD unit itself as the subject of the current IECEX/ATEX certification.

Functionally, the GLD integrates eight channels of metal-oxide semiconductor gas sensors (MQ series), an edge-AI microcontroller/processor (ESP32-S3), and a LoRa radio module (star-topology wireless transmission) within a single fixed-point unit installed at locations with gas-leak risk. An on-device AI gas-classification model runs directly on the unit so that detection decisions do not depend on a continuous connection to a central server. When gas concentration exceeds a defined threshold, the unit triggers a local alarm (an integrated visual/audible alarm module) and simultaneously transmits an alarm notification over the LoRa network to the operator dashboard.

The enclosure is designed for hazardous-area deployment at refinery sites, using metal materials (aluminum alloy and stainless steel) and mounted via a U-bolt mounting plate to existing structures without drilling or welding. Important: this design-intent statement does not constitute a claim that the enclosure has passed testing or has been Ex-certified — the explosion-protection scheme, gas group, temperature class, and target installation zone will be addressed in Section 2.5.

The current production power configuration is continuous 24 VDC, supplied via an AC/DC adapter connected to the site electrical supply. A portable battery power path (Li-ion 18650) remains under development (R&D) and has not become a deployed production configuration.

2.2 • Product Name, Model, and Specification List

Product name	Gas Leak Detector (GLD) — Node Sensor
Model / version	GLD V2
Manufacturer	PT Galaksi Megatama Indonesia
End client / program owner	PT Pertamina Patra Niaga (initial deployment site: Refinery Unit IV, Cilacap)
Primary function	Acquisition of 8-channel gas sensor data and LoRa transmission
Microcontroller	ESP32-S3-WROOM-1U-N16R8
LoRa radio module	E22-900MM22S
Dimensions (L×W×H)	200 × 90 × 290 mm
Enclosure material	Aluminum alloy (ADC12 die-cast) + stainless steel

Source: official product technical datasheet (Revision 4.0), cross-referenced with internal technical specification documentation and EMC parameter measurement data.

2.3 • Functional Description and Technical Parameters (Electrical, Mechanical, etc.)

Functional workflow (normal operating mode): sense → process → transmit. Each of the eight gas sensors continuously samples ambient conditions → data is normalized and classified by the on-device AI model (ESP32-S3) → the result is transmitted over the LoRa network (star-topology transmission) at a configurable interval (default 10 seconds), or immediately (event-driven) when an alarm condition is detected. Gas alarms are triggered through two parallel channels: a local visual/audible alarm module on the unit itself, and a push

notification transmitted over the LoRa network to the dashboard — the alarm-push pathway has been successfully tested on a campus mesh network (field validation at a production refinery installation is still pending).

2.3.a • Electrical Parameters — Node Sensor (GLD)

PARAMETER	SPECIFICATION	STATUS	REMARKS
Main power input	24 VDC	Final — production config.	Continuous power supply from a site AC/DC adapter.
AC/DC adapter input	220 VAC, 50 Hz	Final	Compliant with the Indonesian national electrical grid standard (PLN).
AC/DC adapter output	24 VDC	Final	
Internal operating voltage	5 VDC & 3.3 VDC	Final	5 VDC for the MQ sensor/heater circuitry; 3.3 VDC for ESP32-S3 logic. Per-rail current not yet documented separately.
Equipment boundary (mains exposure)	None — 24 VDC only	Final	The Node Sensor (GLD) itself receives only 24 VDC at its terminal and never carries 220 VAC internally. The AC/DC adapter (220 VAC/50 Hz → 24 VDC) is a separate, site-supplied device outside the physical boundary of the certified unit; its own installation location (e.g., relative to the hazardous-area zone) is a site electrical-design matter, not a parameter of this equipment.
Maximum input current	≈0.33 A @ 24 VDC	Calculated	Calculated from measured maximum power consumption (7.995 W) divided by 24 VDC.
Maximum power consumption	7.995 W @ 24 VDC	Measured — production config.	Applies to the continuous-power configuration. The battery (R&D) configuration is recorded separately at 5.75 W — a different operating mode, not a data conflict.

PARAMETER	SPECIFICATION	STATUS	REMARKS
Electrical protection (fuse, reverse polarity, overvoltage, overcurrent)	2× resettable PPTC fuse (F1/F2); TVS/ESD suppression diodes (D1, D4/D5/D11, D6); Schottky diodes (D7/D13, D8/D9, D12)	Partially available — component-level evidence	Real components exist per the EasyEDA/JLPCB bill of materials (Section 2.6.b): 2× Littelfuse MINISMDC260F/16 resettable fuses (F1, F2, overcurrent), a Ruilong SMBJ33A TVS diode (D1) and 3× UMW LESD5D5.0CT1G / 1× DOWO SM712 ESD-protection arrays (D4, D5, D11, D6), and multiple Schottky diodes — MDD SS54 (D7, D13), MDD SS14 (D8, D9), GOOD-ARK SK36 (D12). This is genuine evidence that overcurrent and transient/ESD protection circuitry exists on the board. The rail and circuit role of each device are now tabulated from the released schematic (Section 2.6.a.7). Two points remain to be confirmed there: the voltage rating of F1/F2 on the 24 V line, and whether the input P-channel MOSFETs are intended to provide reverse-polarity protection.

2.3.b • Sensor and Communication Parameters — Node Sensor (GLD)

PARAMETER	SPECIFICATION	STATUS	REMARKS
Gas sensors	MQ-2, MQ-3B, MQ-4, MQ-5, MQ-6, MQ-7B, MQ-8, MQ-135 (8 channels)	Final	Metal-oxide semiconductor sensors; sensing element directly exposed to ambient air.
AI gas-classification model	On-device classifier — 3 classes: Clean Air, LPG, H ₂	Final	Runs locally on the ESP32-S3 (Running/Inference mode); outputs a class label and a confidence value. Does not yet cover CO ₂ , Benzene, CO, or H ₂ S.
Environmental sensor (temperature/humidity)	SHT40-AD1B-R2 (I2C)	Final	Auxiliary temperature/humidity input included in the primary hardware design; values are available for status/telemetry.
Analog-to-digital converter	ADS1256IDBR	Final	24-bit multi-channel analog acquisition for the 8 sensor channels; 30,000 SPS, firmware SPI clock 1.92 MHz.

PARAMETER	SPECIFICATION	STATUS	REMARKS
Processing unit	ESP32-S3-WROOM-1U-N16R8	Final	Certified under FCC (2AC7Z-ESPS3WROOM1U), TELEC, and CE (per Espressif data) — RF/EMC certifications, not an “Ex component” certification.
Communication module	LoRa, E22-900MM22S module	Final	Certified under CE, FCC, and RoHS (per Ebyte data) — RF/EMC certifications, not an “Ex component” certification.
Operating frequency	920 MHz	Final	Star-topology transmission; within the regional 920–923 MHz ISM band (Indonesia).
Transmit power (firmware configuration)	17 dBm	Final	The radio module supports up to 22 dBm — 17 dBm is an operational configuration, not the module's maximum limit.
Bandwidth / spreading factor / coding rate	125 kHz / SF7 / CR 4/5	Final	Source: EMC parameter table and official product technical datasheet.
Antenna	External, omnidirectional, SMA male connector, 3 dBi gain	Final	On some units, the 2.4 GHz Wi-Fi antenna remains inside the enclosure and must be relocated externally.
Data transmission interval	Configurable, default 10 seconds	Final	Alarm events are transmitted immediately, independent of the periodic interval.
Other interfaces	SPI, LoRa	Final	External ports/connectors: USB, sensor, power, fan, antenna, alarm buzzer.
RS-485 / Modbus interface	Read-only Modbus RTU slave, 9600 bit/s 8N1, Unit ID 1 (THVD1410DR transceiver)	Final	Provides 8 read-only registers (device status word, classification result, confidence, battery voltage, power source, external power, LoRa transmission counter, node ID); not used for product control.

2.3.c • Mechanical Parameters — Node Sensor (GLD)

PARAMETER	SPECIFICATION	STATUS	REMARKS
Enclosure material	Aluminum alloy (ADC12 die-cast) + stainless steel	Final	Per project confirmation, the production enclosure is sourced from a commercially available CE/ATEX-marketed explosion-proof gas-detector housing product line (referenced supplier listing: Alibaba.com, “CE ATEX

PARAMETER	SPECIFICATION	STATUS	REMARKS
			Explosion Proof H2 Sensor”), consistent with the cast-metal housing, threaded “Ex”-marked cable entry, and sensor mesh cover shown in the product photography (Section 2.4). This is corroborated by an internal case CAD drawing (“GLD ATEX CASE v3,” dated 8 September 2026) specifying a cylindrical sensor-case body (Ø102 mm outer housing ring, Ø90/Ø80 mm internal bores) that incorporates a stainless-steel filter mesh disc, a small DC cooling fan, and a transparent viewing window. PVC is not used in any housing or bracket component. Alloy grade — ADC12: the enclosure metal is die-cast aluminum grade ADC12 (a standard, widely used die-cast aluminum alloy), per project confirmation. Caveat on how this grade was established: ADC12 is the material called out on a Chinese enclosure-component manufacturer's drawing for a related die-cast sub-component (“Universal Base,” part no. BP18-1Z, from the same casing-partner supply chain — see Section 2.6.a); the “GLD ATEX CASE v3” drawing of the main enclosure body itself does not carry its own material callout. The project has confirmed ADC12 as the enclosure grade on the basis that the same casing manufacturer's die-cast product line uses this alloy consistently — this is an extrapolation from the supplier's sub-component drawing, not a direct material callout on the main enclosure drawing, and should be verified against the casing partner's own datasheet for the enclosure body before formal ExCB submission.

PARAMETER	SPECIFICATION	STATUS	REMARKS
Dimensions (L × W × H)	200 × 90 × 290 mm	Final	Consistent between the technical specification documentation and the EMC parameter table.
Total weight	2.378 kg	Final	Total weight of PCB, casing, and sensors combined; project-confirmed value, not yet documented against a calibrated weighing record.
Mounting method	U-bolt mounting plate, following the design already installed at the refinery	Final	Mounted to existing structures without drilling or welding.
Ingress protection (IP rating)	IP66	Final	Project-confirmed value; not yet independently tested/verified against IEC 60529.
Cable entry (gland)	M20×1.5 cable gland (IP66-rated); power cable is 2-conductor, labeled L+/L- (positive/negative of the 24 VDC supply), each conductor ≈0.75 mm diameter	Final	Conductor count and per-conductor wire diameter confirmed by the project. Gland thread size (M20×1.5) is selected by engineering judgment to match the BP18-1Z reference drawing from the same casing-manufacturer supply chain (Section 2.6.a) — a standard, IP66-rated gland size well suited to this conductor count/diameter, not an independent gland datasheet or fit test on GLD's own enclosure. Overall cable outer diameter and insulation rating remain to be documented.
Antenna mounting	External, SMA male connector	Final	
Operating temperature	–20°C to +60°C (ambient)	Final	Project-confirmed value — a key input for temperature class (T1–T6) determination in Section 2.5. Not yet documented against a calibrated test record.
Operating humidity	5–95% RH, non-condensing	Final	Project-confirmed value. Not yet documented against a calibrated test record.

Source: internal technical specification documentation, Sections 1.1–1.3 (Node Sensor), cross-referenced with the official product technical datasheet (Revision 4.0) and component certification verification (ESP32-S3-WROOM-1U, E22-900MM22S).

Rows marked “Pending confirmation” above do not reflect a documentation oversight — this is an honest status indicator. These fields are intentionally left blank because no official data yet exists (not yet measured, tested, or

decided). They must not be filled with estimates in future revisions without a clear supporting data source.

2.4 • Product Photographs — Overall and Key Components

The seven photographs below were taken directly from the Node Sensor (GLD) V2 prototype unit (no capture-date metadata is available in the source files). The first three photographs show the fully assembled unit from different angles; the remaining four show key components in close-up.



Complete unit — front view

The fully assembled Node Sensor (GLD): the main enclosure (blue) with sensor mesh cover, external LoRa antenna, and a visual alarm module (cylindrical housing, red lens) mounted on the right side.



Complete unit — alternate angle

An alternate view of the assembled unit, showing the position of the antenna, alarm module, and the cable-gland side of the enclosure.



Complete unit — front view (without antenna and alarm module)

The main enclosure with the sensor mesh cover installed, shown in a flat front view illustrating the enclosure shape and its four mounting bolt holes.



Key component — PCB inside enclosure (before sensor modules)

The main PCB installed inside the enclosure prior to gas sensor module installation. The ESP32-S3-WROOM-1U microcontroller, LoRa radio module, and power circuitry are visible.



Key component — PCB with eight gas sensor modules

The same PCB with eight channels of MQ-series gas sensors (MQ-2, MQ-3B, MQ-4, MQ-5, MQ-6, MQ-7B, MQ-8, MQ-135) installed around the central microcontroller.



Key component — visual alarm module (close-up)

The visual alarm (beacon/strobe) module, constructed of metal with a red lens, connected to the main enclosure via a dedicated cable gland.



Key component — sensor mesh cover (close-up)

The metal mesh cover that protects the gas sensing elements while serving as a diffusion path for gas to reach the sensors beneath.

Photographic completeness — status as-is. The seven photographs above cover the complete unit and its key components (PCB, sensor modules, alarm module, mesh cover), substantively satisfying checklist Item 2.4. Still outstanding: (a) formally labeled photographs of each face (front/back/left/right/top/bottom) with a scale reference, as is customary in ExCB submission packages; (b) separate photographs of individual components such as the battery, gaskets/seals, terminals, and cable glands.

2.5 • Description of Intended Use and Installation Environment (Gas Group, Temperature Class, Area Classification)

The GLD is intended for continuous, fixed-point monitoring of flammable and toxic gas leaks in oil & gas refinery hazardous areas — process units, tank farms, pipe racks, and loading/unloading areas. The specific Ex

classification parameters below reflect the project engineering team's current recommendation, developed from the intended deployment envelope and the gases the device is designed to detect; none of these parameters has yet been confirmed with the certification body (ExCB) or a notified body, and none should be treated as final.

PARAMETER	RECOMMENDED VALUE	STATUS	RATIONALE / REMARKS
Gas group	IIC	Team recommendation — pending ExCB confirmation	Hydrogen (H ₂) is one of the three gases the classifier is designed to detect (Section 2.3.b); IIC is required for hydrogen and inherently covers IIB and IIA.
Temperature class	T4 (≤135°C)	Team recommendation — pending verification	Supported by a preliminary internal surface-temperature measurement (Section 2.6.f): the highest value, extrapolated to the +60°C maximum ambient, is ≈114°C (MQ sensor body, fan-stalled condition), below the T4 limit less the 5 K margin (130°C). Not yet final — several measurement points and conditions remain open (see 2.6.f), and confirmation requires ExCB type testing.
Ambient temperature range	–20°C to +60°C	Final	Project-confirmed value, consistent with the Mechanical Parameters table (Section 2.3.c). Used as the reference for extrapolating the preliminary surface-temperature measurements in Section 2.6.f to worst-case ambient.
Area classification (zone)	Zone 1	Team recommendation — pending ExCB confirmation	Assessed as sufficient for the general refinery deployment envelope, including areas near storage. Zone 0 would only apply if the detector were installed directly inside a tank vapor space.

PARAMETER	RECOMMENDED VALUE	STATUS	RATIONALE / REMARKS
Type of protection	Ex d — flameproof enclosure	Team recommendation — pending ExCB confirmation	Selected over Ex e and Ex i based on the existing enclosure design: a die-cast aluminum (ADC12) housing with a stainless-steel wire mesh over the sensing element functions as a flame path/flame arrestor, the same pattern used by the three comparable certified gas detectors reviewed as references (all Ex d). Ex i was set aside because the MQ-series sensor heaters draw continuous power (≈ 0.75 W each) directly from an unregulated 24 VDC supply rather than through certified current-limiting barriers, difficult to reconcile with intrinsically-safe energy limits for Group IIC. Ex e was not pursued because it is not intended for enclosures with normally-energized, potentially arcing parts such as the sensor heaters.

These are engineering recommendations, not a certification decision. An independent readiness assessment of the same checklist item, prepared separately from the underlying firmware/hardware repository, reached the same conclusion — gas group, temperature class, ambient range, zone, and type of protection are not yet formally established. Final classification requires explicit agreement with the ExCB.

2.6 • Design and Manufacturing Information

Nine sub-items (a–i) per the original checklist. Status is reported item by item below; most sub-items are not yet available — this is reported plainly rather than implied to be complete.

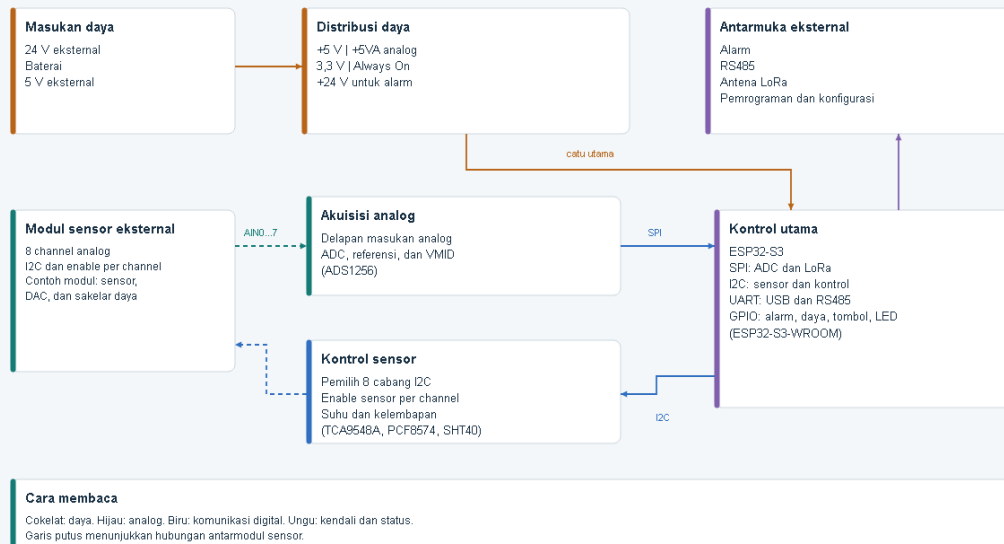
2.6.a • Complete Drawings (Assembly, Component, Electrical Schematic, PCB Layout, Enclosure Structure, Junction Box, Terminal, Grounding)

An electrical schematic capture and a corresponding PCB layout exist for the GLD V2 main board as native EasyEDA/JLCPCB source design files (not just a derived summary). From the schematic's traced net list, a supporting block-diagram set (9 sheets, functional/block level, 204 components mapped with documented pin-to-net traceability) has been produced and is reproduced in full below. Field labels in the source diagrams are in Indonesian; English captions are provided under each sheet.

PERTAMINA GLD / SCHEMATIC BLOCK DIAGRAM

01 Arsitektur keseluruhan

Catu daya, akuisisi sensor, kendali, dan komunikasi dalam satu perangkat.



01 / 9

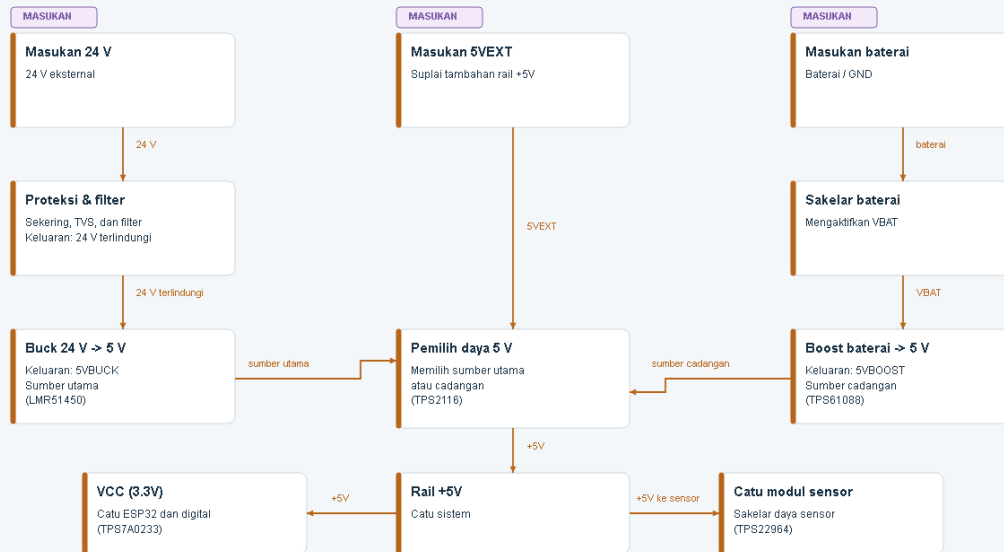
Sheet 1 of 9 — Overall architecture

Power input, power distribution, analog acquisition, sensor control, main control (ESP32-S3), and external interfaces (alarm, RS-485, LoRa antenna, programming/configuration).

PERTAMINA GLD / SCHEMATIC BLOCK DIAGRAM

02 Distribusi daya utama

Tiga masukan daya, pemilihan sumber 5 V, rail analog, dan catu sensor.



24 V terlindungi: keluaran setelah sekering, TVS, dan filter.

02 / 9

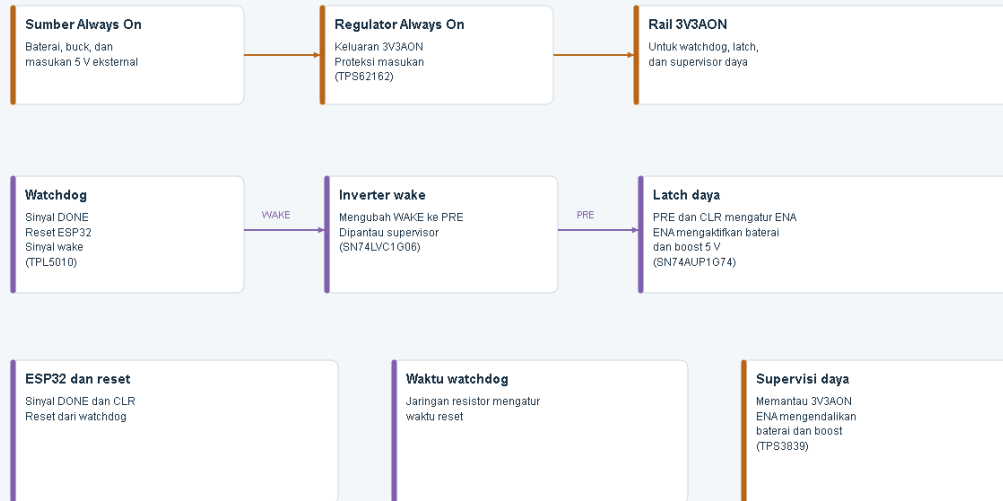
Sheet 2 of 9 — Main power

Main power input rails and distribution: +5 V / +5VA analog, 3.3 V, and +24 V for the alarm circuit.

PERTAMINA GLD / SCHEMATIC BLOCK DIAGRAM

03 Always On, watchdog, dan latch

Always On (AON) adalah rail daya yang tetap aktif untuk menjaga pengawasan dan pengendalian daya.



03 / 9

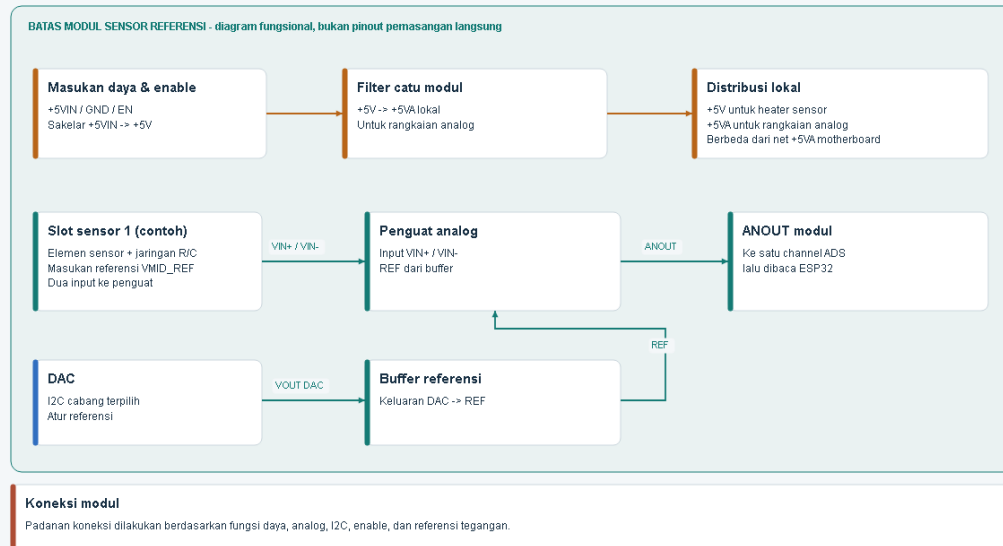
Sheet 3 of 9 — Always-on power

The always-on power domain that remains active independent of the main control state.

PERTAMINA GLD / SCHEMATIC BLOCK DIAGRAM

04 Modul sensor: jalur fungsional

Setiap modul menerima daya, I2C, enable, dan menyediakan satu keluaran analog.



04 / 9

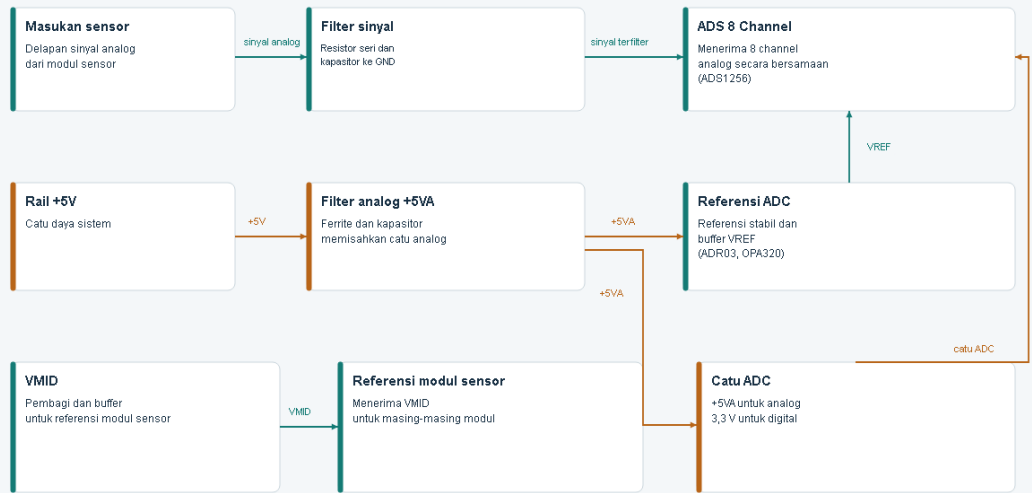
Sheet 4 of 9 — Sensor module

The external gas sensor module: 8 analog channels with per-channel I2C enable, including the sensor, DAC, and power-switch sub-blocks.

PERTAMINA GLD / SCHEMATIC BLOCK DIAGRAM

05 Akuisisi analog dan tegangan referensi

Jalur analog memiliki filter catu terpisah, referensi tegangan, VMID, dan ADS 8 Channel.



05 / 9

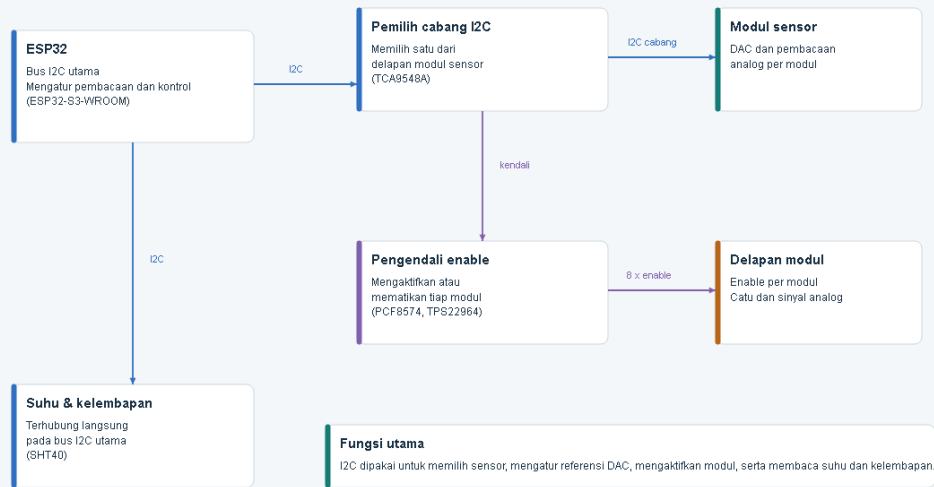
Sheet 5 of 9 — Analog acquisition

The 8-channel analog acquisition front end: analog inputs, ADC (ADS1256), reference, and VMID biasing.

PERTAMINA GLD / SCHEMATIC BLOCK DIAGRAM

06 I2C dan kendali sensor

I2C memilih cabang sensor, mengatur DAC, dan mengendalikan daya tiap modul.



06 / 9

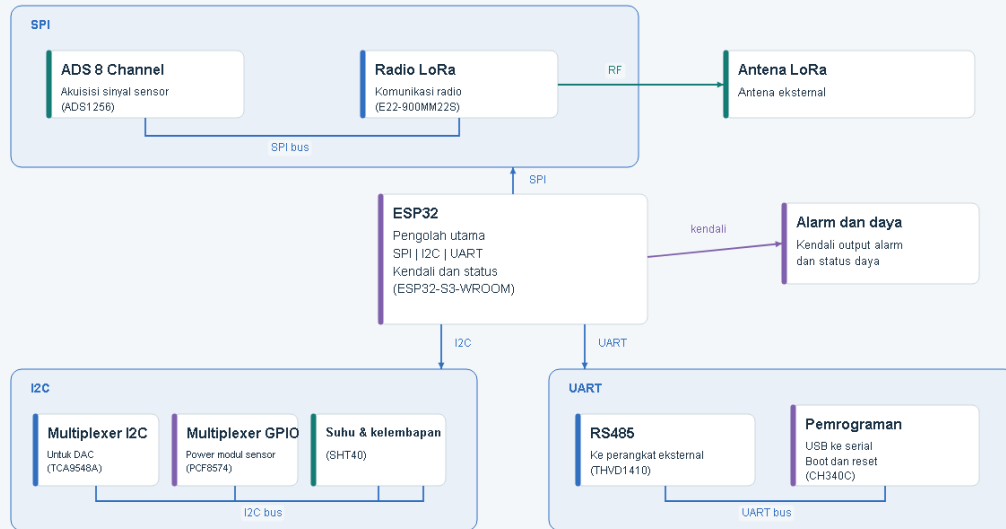
Sheet 6 of 9 — I2C & sensor control

I2C channel selection (TCA9548A), per-channel sensor enable (PCF8574), and the temperature/humidity sensor (SHT40).

PERTAMINA GLD / SCHEMATIC BLOCK DIAGRAM

07 Komunikasi dan koneksi ESP32

Koneksi dikelompokkan menurut bus SPI, I2C, dan UART, kendali perangkat dipisahkan dari komunikasi.



07 / 9

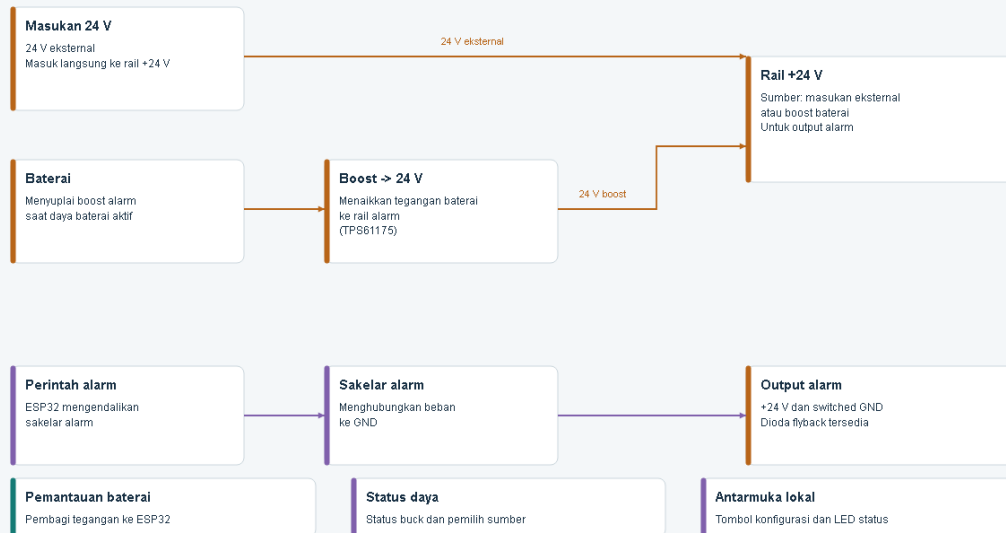
Sheet 7 of 9 — ESP32 connections

Main controller (ESP32-S3) interface map: SPI to the ADC, I2C to sensor/control, UART for USB and RS-485, and GPIO for alarm, power, button, and LED.

PERTAMINA GLD / SCHEMATIC BLOCK DIAGRAM

08 Alarm, baterai, dan status daya

Rail +24 V alarm dapat berasal dari masukan 24 V eksternal atau boost baterai.



08 / 9

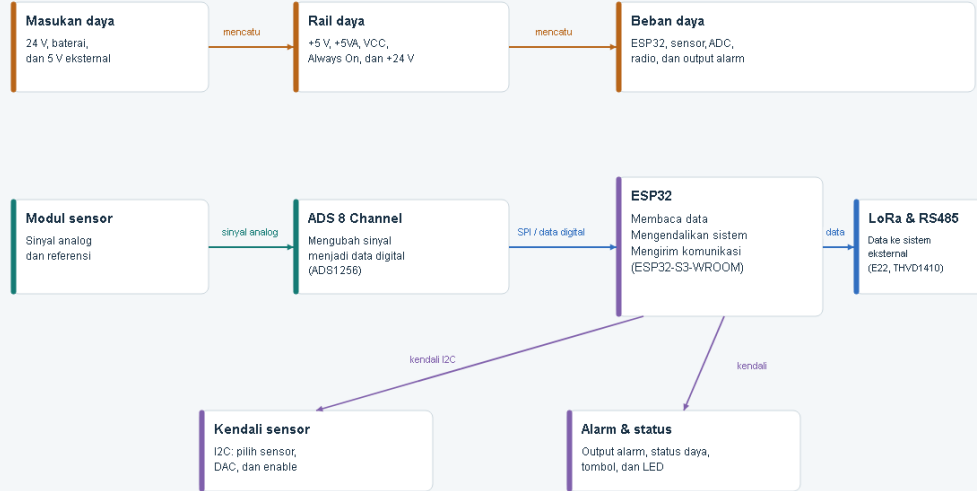
Sheet 8 of 9 — Alarm & status

Alarm output and status-indication circuitry driven from the main controller.

PERTAMINA GLD / SCHEMATIC BLOCK DIAGRAM

09 Alur energi, data, dan kendali

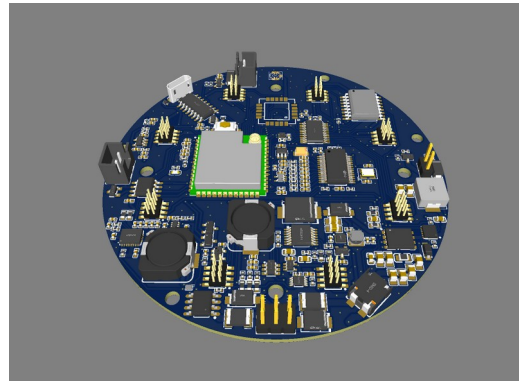
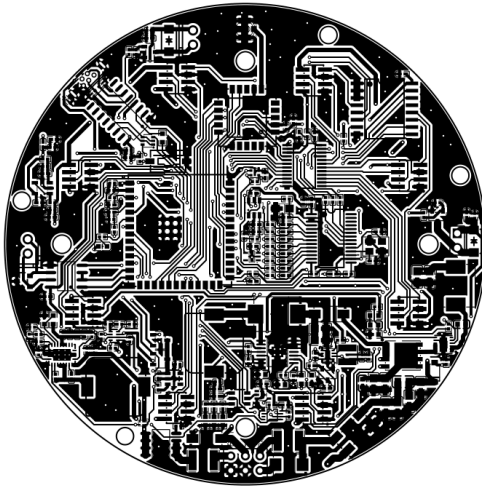
Daya mencatu perangkat; data sensor mengalir ke ESP32; ESP32 mengendalikan komunikasi dan keluaran.



09 / 9

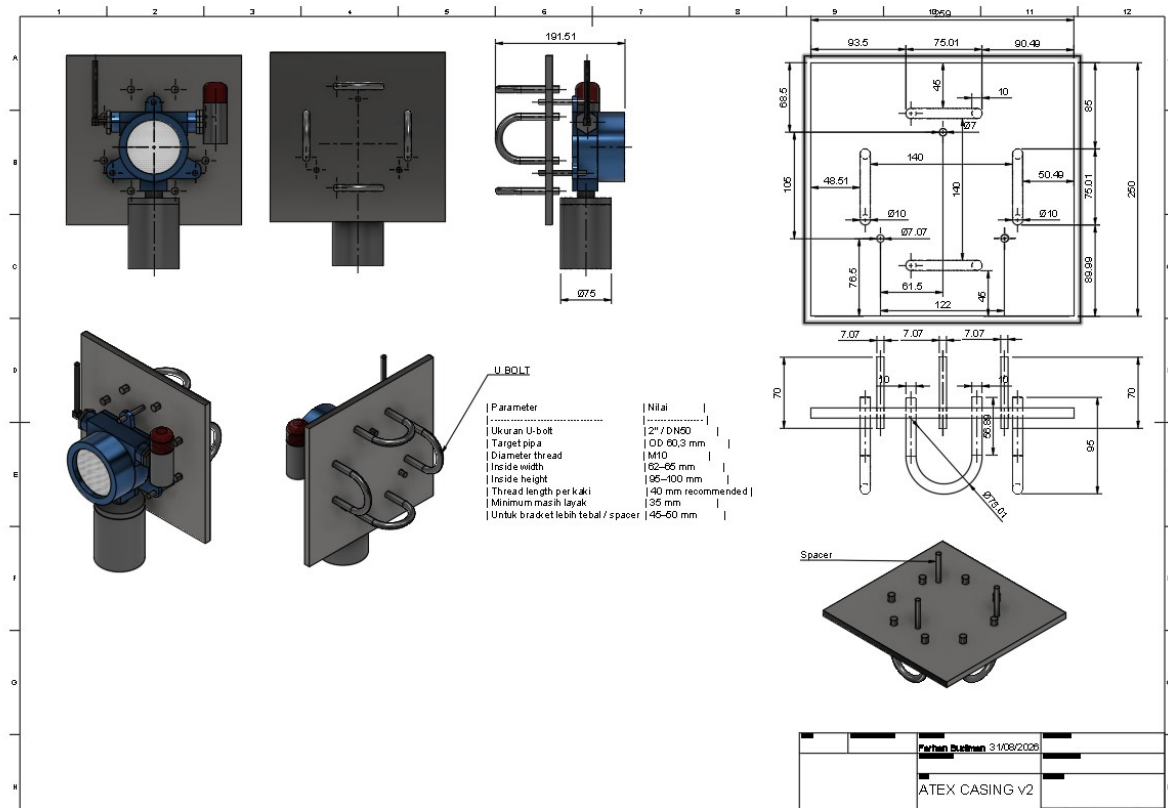
Sheet 9 of 9 — System flow

Overall signal and data flow across the board, summarizing how the preceding eight sheets connect end to end.

**Main board PCB layout (top copper) and 3D populated render**

Both exported directly from the EasyEDA/JLCPCB source project (production-intent board, circular outline with six mounting holes). The 3D render shows the ESP32-S3-WROOM module, micro-USB connector, power inductors, and 8-channel I2C header block in their real placed positions; it is an illustrative render, not a dimensioned drawing.

A dimensioned mechanical drawing sheet also exists for the enclosure's external envelope and its mounting hardware, drafted from a solid CAD model (STEP format, millimeter units) with a formal title block, orthographic and isometric views, and a parts table — reproduced below.



Enclosure envelope & mounting bracket — dimensioned drawing sheet

Orthographic and isometric views with real dimensions (enclosure neck Ø75 mm, overall probe height 191.51 mm, mounting plate 250×250 mm with toleranced hole pattern), a U-bolt parameter table (2"/DN50, M10 thread), and a title block (drafted 31 Aug 2026). Source: dimensioned CAD drawing derived from a STEP solid model of the enclosure and bracket assembly.

DRAWING TYPE	STATUS	REMARKS
Electrical schematic (component-level) — main board	Available	Actual component-level schematic issued as 26 sheets (drawing nos. PGLD-GLD-V4-MB-SCH-01 to -26, rev. 1.0, 19 Jul 2026), each with a sheet information table and component description table (Section 2.6.a.7). Drawing-number prefix to be aligned with the applicant's document control; verification items listed in 2.6.a.7 remain open. Sensor-module board schematic not yet included.
PCB layout — main board	Partially available	Ten annotated layout views (top, bottom, placement, dimensions Ø84 mm, mounting holes, connectors, power area, sensor interface, grounding) issued as PGLD-GLD-V4-MB-PCB-01 to -10 (Section 2.6.a.8). Open: layout revision/date, laminate data (grade, UL 94, Tg, CTI, thickness, copper weight) and the sensor-module board layout.

DRAWING TYPE	STATUS	REMARKS
Assembly drawing	Partially available	A dimensioned drawing sheet exists for the enclosure/bracket mounting assembly (title block, orthographic + isometric views, parts table). It covers the external envelope and mounting hardware, not the internal PCB/component assembly sequence.
Component drawing	Not yet available	
Enclosure structure drawing (gap, length, volume)	Partially available	External envelope is dimensioned (from a real STEP solid model, millimeter units) in the drawing above. The Ex-d-specific flame-path parameters — joint gap, length, and free internal volume — are not called out; that dimensioning has to be added deliberately once a protection concept is confirmed (Section 2.5), not derived automatically from the CAD export.
Junction box / terminal / grounding connection drawings	Candidate reference only	A Chinese enclosure-component supplier's drawing for a die-cast "Universal Base" (part no. BP18-1Z, material ADC12, 0.6 kg, drawing scale 1:1) has been received via the casing partner's supply chain. It specifies 2× M20×1.5-6H cable-entry threads, 4× M4-6H and 1× M5×1.5-6g fixing threads, 3× Ø8 mm mounting holes, an internal bore up to Ø94.7 mm, and overall dimensions approximately 124×115×67.5 mm. Its footprint does not match GLD's main enclosure (200×90×290 mm), so this is understood as a candidate junction-box/terminal sub-component under evaluation, not yet confirmed as part of GLD's design. A dedicated grounding connection drawing has not been provided.

Real dimensioned CAD drawings now exist for the enclosure's external envelope and mounting hardware, and native PCB layout/schematic source files exist for the electronics. What is still missing for a certifiable drawing package: internal component/assembly drawings, junction box and grounding-connection drawings, and — specific to explosion protection — flame-path gap/length/volume dimensioning, which requires a confirmed protection concept before it can be drawn.

2.6.a.7 Main PCB schematic

The electrical schematic of the GLD main board is provided as the actual circuit schematic (not a block diagram). It is presented as 26 figures, each cut from the original EasyEDA schematic around one functional block. Every component in a figure is enclosed in a numbered boundary box, and the same number is used in the description table printed with that figure. Each figure carries a sheet information table with the drawing/document number, revision, date, title and sheet number (n of 26).

Sheet register

Drawing no.	Title	Requirement covered
PGLD-GLD-V4-MB-SCH-01	24 VDC input, input protection and 24 V to 5 V buck (U36)	24 VDC input; input protection
PGLD-GLD-V4-MB-SCH-02	Power connector J1, battery input and external 5 V	24 VDC input
PGLD-GLD-V4-MB-SCH-03	Battery load switch (U14)	Input protection
PGLD-GLD-V4-MB-SCH-04	5 V power-path mux (U13)	5 V rail
PGLD-GLD-V4-MB-SCH-05	Battery boost to about 24 V (U1, TPS61175)	5 V rail / +24 V backup
PGLD-GLD-V4-MB-SCH-06	Battery boost to 5 V (U41, TPS61088)	5 V rail
PGLD-GLD-V4-MB-SCH-07	5 V to 3.3 V buck (U43)	3.3 V rail
PGLD-GLD-V4-MB-SCH-08	Always-on 3.3 V LDO and its source selection (U42)	3.3 V rail
PGLD-GLD-V4-MB-SCH-09	TPL5010 timer/watchdog (U46)	3.3 V rail
PGLD-GLD-V4-MB-SCH-10	ENA latch (U51) and open-drain inverter (U15)	3.3 V rail
PGLD-GLD-V4-MB-SCH-11	3V3AON voltage supervisor (U16)	3.3 V rail
PGLD-GLD-V4-MB-SCH-12	Sensor module connector H2 (one connector as example)	MQ heater supply
PGLD-GLD-V4-MB-SCH-13	Sensor module connectors H1–H8	MQ heater supply; sensor interface
PGLD-GLD-V4-MB-SCH-14	Sensor module enable I/O expander (U12)	MQ heater supply (module enable)
PGLD-GLD-V4-MB-SCH-15	VMID bias of about 2.5 V (U3)	MQ sensor analog circuitry
PGLD-GLD-V4-MB-SCH-16	Analog +5VA supply filter (L6)	MQ sensor analog circuitry
PGLD-GLD-V4-MB-SCH-17	Sensor module I2C multiplexer (U33)	MQ sensor analog circuitry
PGLD-GLD-V4-MB-SCH-18	Temperature and humidity sensor (U44, SHT40)	Analog circuitry (environment)
PGLD-GLD-V4-MB-SCH-19	ADS1256 ADC with input filters (AIN), clock and decoupling	ADC; analog circuitry
PGLD-GLD-V4-MB-SCH-20	2.5 V voltage reference and VREF buffer	ADC
PGLD-GLD-V4-MB-SCH-21	ESP32-S3-WROOM-1U-N16R8 with support parts (incl. status LED)	ESP32-S3; LED
PGLD-GLD-V4-MB-SCH-22	USB, CH340C and auto-reset	ESP32-S3 (programming)
PGLD-GLD-V4-MB-SCH-23	E22-900MM22S LoRa module and antenna connector	LoRa
PGLD-GLD-V4-MB-SCH-24	5 V fan driver (Q5, J3)	Fan driver
PGLD-GLD-V4-MB-SCH-25	24 V alarm output driver (Q4, J2)	LED / buzzer (alarm) driver
PGLD-GLD-V4-MB-SCH-26	RS-485 transceiver (U47)	RS-485 / Modbus

All sheets: revision 1.0, date 2026-07-19.

Coverage of the required schematic content

Required content (2.6.a.7)	Where shown	Status
24 VDC input	Sheets 01, 02	Provided

Required content (2.6.a.7)	Where shown	Status
Input protection	Sheets 01–03; protection-device table below	Provided — see findings 1–2
5 V rail	Sheets 01 (buck U36), 04–06	Provided
3.3 V rail	Sheets 07–11	Provided
MQ heater supply	Sheets 12–14 (+5 V and enable to each sensor module)	Partly — heater switch is on the sensor module (finding 4)
MQ sensor bridge / analog circuitry	Sheets 15–18; AIN filters on sheet 19	Partly — load resistor is on the sensor module (finding 4)
ADC	Sheets 19, 20	Provided
ESP32-S3	Sheets 21, 22	Provided
LoRa	Sheet 23	Provided
Fan driver	Sheet 24	Provided
LED / buzzer driver	Sheet 25 (24 V alarm output J2); status LED on sheet 21	Provided
RS-485 / Modbus	Sheet 26	Provided
Protection devices	Table below	Provided
Grounding	Text below	Provided — PE point not drawn (finding 5)
Title block per sheet	Sheet information table on every figure	Provided

Protection devices

Designator	Part	Location	Function	Sheet
F1	MINISMDC260F/16-2	Series with 24V+ after J1	Overcurrent on 24 V line (PTC)	01
D1	SMBJ33A	After F1, to 24V–	Surge / transient clamp on 24 V line (TVS)	01
L1	ACM7060-301-2PL	Series with 24V+ and return	Common-mode noise	01
L4	MPZ2012S101AT000	Series with +24V	High-frequency noise (ferrite)	01
D2 / R26	LBZT52C12T1G / 100k	Gate of Q3	12 V gate-source limit for Q3	01
Q3	SI7465DP-T1-GE3	+24V to buck input	High-side switch	01
F2	MINISMDC260F/16-2	Series with battery input J1.2	Battery overcurrent (PTC)	02
Q2 / R1	AO4407 / 100k	Battery to VBAT_IN	High-side switch	02
D7 / D13	SS54	5VEXT to +5V; boost to +24V	Reverse-current blocking (OR-ing)	02, 05
D3 / D10	BAT54C	Source of U42 (3V3AON)	OR-ing of supply sources	08
D8 / D9	SS14	Loads J2 and J3	Flyback for inductive loads	24, 25
D4 / D5 / D11	LES5D5.0CT1G	USB D–, D+, VBUS	USB ESD	22
D6	SM712	RS-485 A/B	RS-485 line TVS	26
L6	GZ1005D600TF	+5V to +5VA	Analog supply noise isolation	16

Grounding

- The circuit uses a single GND net; there is no separate AGND or chassis-ground net. The AGND/PGND/DGND pins of all ICs are connected to GND.
- The 24 V return (J1 pin 5, 24V–) enters GND through one winding of common-mode choke L1 (pin 2 to pin 3). J1 pins 1 and 3 are connected directly to GND.
- The analog supply +5VA is separated from +5V by ferrite bead L6; analog and digital ground are joined at GND.
- The shell/EP pads of USB1 and the GND pads of the LoRa module/antenna connector are connected to GND.

- A connection point from circuit GND to the enclosure / protective earth is not drawn on the schematic; it is defined by the enclosure grounding/bonding drawing.

Findings to be resolved before the dossier is finalised

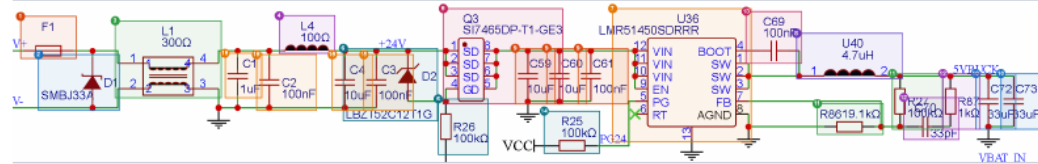
#	Finding (from the schematic)	Action
1	F1 and F2 are MINISMDC260F/16-2; the "/16" suffix normally means a 16 V rating, while F1 is on the 24 V line.	Confirm the PTC voltage rating from the datasheet or change the part.
2	Q3 (24 V) and Q2 (battery) have the source on the input side and the gate pulled to GND; with this orientation the body diode does not block reverse polarity.	Confirm the intended function. Any description of "P-channel MOSFET reverse-polarity protection" must match the actual circuit.
3	Battery input, battery boost to 24 V (TPS61175) and to 5 V (TPS61088), battery load switch and power-path mux are populated on the main board.	Decide whether the battery path is excluded (depopulated / not fitted) or included in the certified configuration; the production configuration is stated as 24 VDC only.
4	Each sensor connector H1–H8 carries only +5 V, VMID, enable, I2C and AIN. The per-channel heater switch and the sensor load resistor are not on the main board.	Provide the sensor-module schematic and layout as a separate drawing set.
5	Single GND net; no chassis/PE connection point on the board.	Show the GND-to-enclosure/PE arrangement in the grounding/bonding drawing.
6	RS-485 has TVS D6 only; no 120 Ω termination or A/B bias resistors.	Confirm termination/bias is provided externally (installation note) or is not required.
7	Converter output voltages (≈ 5.0 V, ≈ 24 V, ≈ 5.3 V) are calculated from feedback resistors, not measured.	Record measured rail voltages in the routine/functional test.
8	The load type on alarm output J2 (24 V siren/beacon) is not defined on the schematic.	Identify the alarm module (manufacturer/model, current) in the BOM (6.b).

Schematic figures (sheets 01–26)

The pages below reproduce the schematic figures exactly as issued, including the per-figure sheet information table and component description table.

24 VDC input

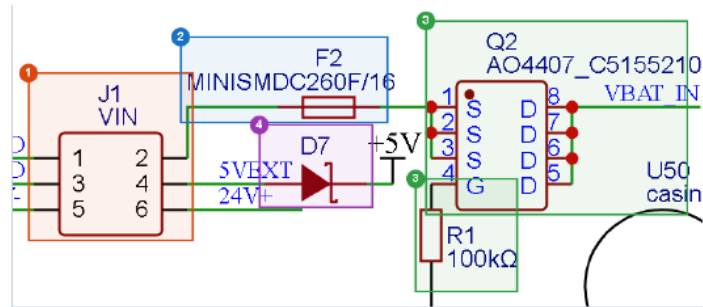
Figure 1 - 24 VDC input, input protection and 24 V to 5 V buck (U36)



Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-01	1.0	2026-07-19	24 VDC input, input protection and 24 V to 5 V buck (U36)	1 of 26

Block function: The 24 VDC path from the connector to the 24 V to 5 V buck (U36). **Requirement covered:** 24 VDC input and input protection

No	Component	What it is	What it does
1	F1	Resettable PTC fuse MINISMDC260F/16-2 (2.6 Ahold).	Opens on a short circuit or overload on the 24 V line and recovers by itself. Note: the /16 suffix in the part number usually means a 16 V voltage rating - to be confirmed against the datasheet because this line is 24 V.
2	D1	TVS diode SMBJ33A(33 V standoff).	Clips voltage surges/transients between the 24 V line after F1 and 24V-, so the circuitry behind it is not damaged.
3	L1	Common-mode choke ACM7060-301-2PL (4 pins).	Suppresses common-mode noise on the 24V+ / 24V- pair. Winding 1-4 carries 24V+, winding 2-3 carries the return side (24V- to GND).
4	L4	Ferrite bead MPZ2012S101A(100 ohm @ 100 MHz).	High-frequency noise filter in series with the +24V line before it enters the internal rail.
5	Q3	P-channel MOSFET SI7465DP (source on +24V, drain toward the buck input).	Passes +24V to the 5 V converter. Its source is on the +24V side and its drain on the load side; this orientation should be verified if the part is meant to provide reverse-polarity protection.
6	D2, R26	12 V zener (LBZT52C12) and a 100k resistor to GND.	R26 pulls the gate to GND so Q3 turns on. D2 limits the gate-source voltage of Q3 to 12 V so the gate is not damaged by 24 V.
7	U36	Synchronous buck IC LMR51450 (VIN up to 36 V).	Step-down converter from 24 V to 5VBUCK. EN is tied to VIN (always on); the PG (power good) pin goes to the ESP32 as signal PG24.
8	U40	4.7 uH inductor (GSSM06304R7M2AU).	Buck energy-storage inductor between the SW node and 5VBUCK.
9	C59, C60, C61	Input capacitors 10 uF/50 V x2 + 100 nF.	Supply the pulsed input current of the buck.
10	C72, C73	Output capacitors 33 uF x2 + 10 uF.	Smooth the 5VBUCK output voltage.
11	R27, R86	FB voltage divider 100k / 19.1k.	Sets the output: $0.8 \text{ V} \times (1 + 100/19.1) = \text{about } 5.0 \text{ V}$ (calculated from the resistor values).
12	R87, C70	1k resistor in series with a 33 pF capacitor on FB.	Feed-forward network for stability / transient response.
13	C69	100 nF BOOT-SW capacitor.	Bootstrap for the internal high-side MOSFET driver.
14	R25	100k pull-up to VCC on PG24.	Gives a 3.3 V logic level for the power-good signal read by the ESP32 (IO47).
15	C1, C2, C3, C4	Decoupling capacitors 1 uF, 100 nF, 100 nF, 10 uF/50 V.	Filter ripple/noise and hold up small transients on the +24V rail (paired with L1/L4).

Figure 2 - Power connector J1, battery input and external 5 V

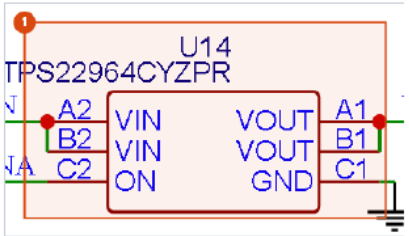
Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-02	1.0	2026-07-19	Power connector J1, battery input and external 5 V	2 of 26

Block function: Power connector, battery input protection and switch, and the external 5 V path. **Requirement covered:** 24 VDC input and input protection

No	Component	What it is	What it does
1	J1	Power connector, 2x3 pins (J1).	The only power entry from outside: pin 6 = 24V+, pin 5 = 24V-, pin 4 = 5VEXT (external 5 V), pins 3 and 1 = GND, pin 2 = battery input (goes to F2).
2	F2	PTC fuse MINISMDC260F/16-2.	Overcurrent protection for the battery input at J1 pin 2.
3	Q2, R1	P-MOSFET AO4407 with a 100k gate resistor to GND.	Passes the battery to net VBAT_IN. R1 turns Q2 on through the gate to GND. The source-on-battery orientation should be verified if the part is meant to provide reverse-polarity protection.
4	D7	SS54 Schottky diode from 5VEXT to +5V.	Lets external 5 V (J1 pin 4) supply the +5V rail without back-feeding.

Input protection

Figure 3 - Battery load switch (U14)



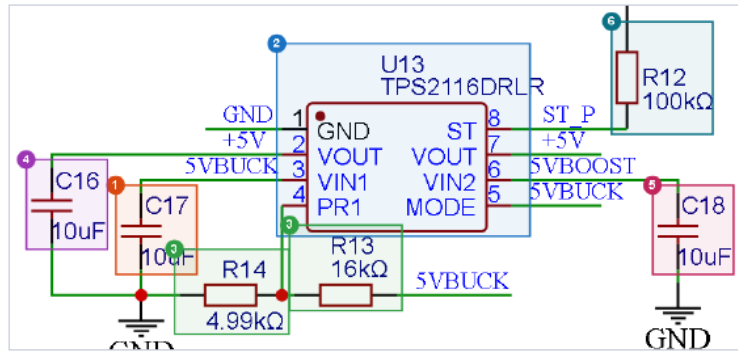
Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-03	1.0	2026-07-19	Battery load switch (U14)	3 of 26

Block function: Battery load switch to the VBAT rail. **Requirement covered:** Input protection

No	Component	What it is	What it does
1	U14	Load switch TPS22964C (VIN = VBAT_IN, VOUT = VBAT, ON = ENA).	Connects/disconnects the battery to the whole VBAT rail (boost converters) according to the ENAsignal from the power-latch circuit.

5 V rail

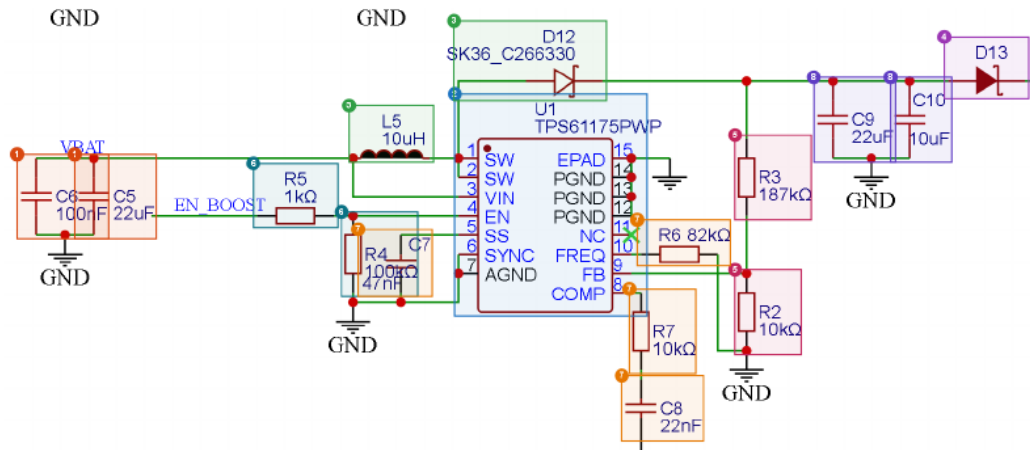
Figure 4 - 5 V power-path mux (U13)



Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-04	1.0	2026-07-19	5 V power-path mux (U13)	4 of 26

Block function: Selects the 5 V source between 5VBUCK and 5VBOOST. **Requirement covered:** 5 V rail

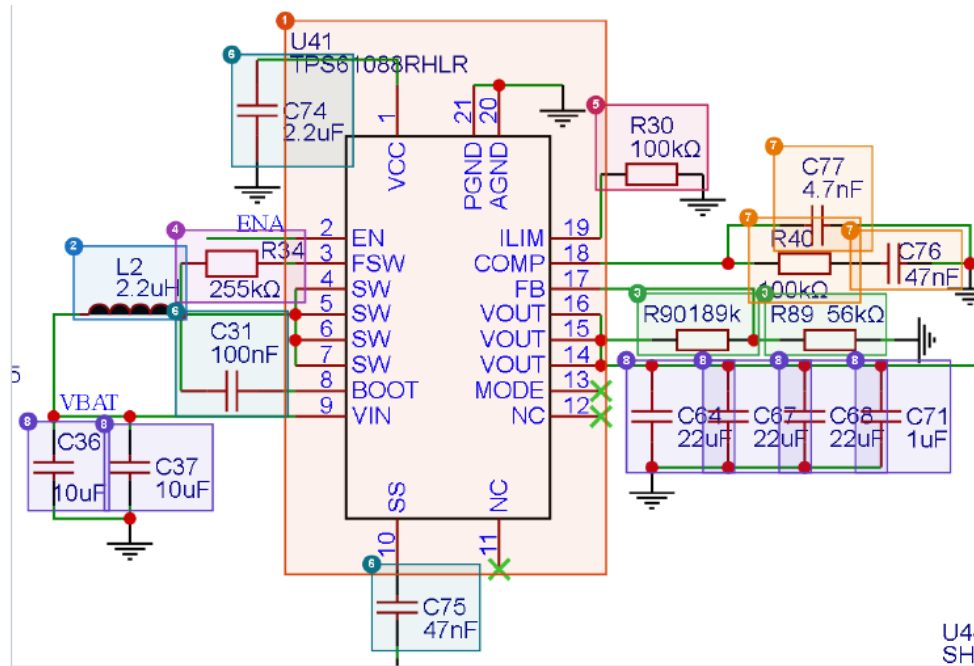
No	Component	What it is	What it does
1	C17	Output capacitors 33 uF x2 + 10 uF.	Smooth the 5VBUCK output voltage.
2	U13	Power mux TPS2116 (VIN1 = 5VBUCK, VIN2 = 5VBOOST, VOUT = +5V).	Automatically selects the 5 V source with priority on 5VBUCK (MODE is tied to 5VBUCK) and switches to 5VBOOST when 5VBUCK is lost. The ST pin goes to the ESP32 as ST_P (source status).
3	R13, R14	16k / 4.99k divider on PR1.	Sets the voltage threshold at which 5VBUCK is considered valid before the mux gives it priority.
4	C16	10 uF and 100 nF capacitors on +5V.	Decoupling of the +5V rail.
5	C18	VBAT input and 5VBOOST output capacitors.	Decoupling on the input and output sides of the boost.
6	R12	100k pull-up to VCC on ST_P.	Gives a 3.3 V logic level for the mux status signal read by the ESP32 (IO18).

Figure 5 - Battery boost to about 24 V (U1, TPS61175)

Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-05	1.0	2026-07-19	Battery boost to about 24 V (U1, TPS61175)	5 of 26

Block function: Boosts the battery to about 24 V as a backup of the +24V rail. **Requirement covered:** 5 V rail / backup of the +24V rail

No	Component	What it is	What it does
1	C5, C6	22 uF and 100 nF capacitors on VBAT.	Decoupling on the load-switch output.
2	U1	Boost converter TPS61175 (VIN = VBAT).	Boosts the battery voltage to about 24 V. Enabled by the EN_BOOST signal from the ESP32 (IO15).
3	L5, D12	10 uH inductor and SK36 Schottky diode.	Boost inductor and rectifier diode from the SW node to the boost output.
4	D13	SS54 Schottky diode from the boost output to +24V.	ORs the boost output onto the +24V rail without back-feeding into the boost converter.
5	R3, R2	FB divider 187k / 10k.	Sets the output to about $1.229 \text{ V} \times (1 + 187/10) = 24 \text{ V}$ (calculated from the resistor values).
6	R4, R5	100k pull-down and 1k series resistor on EN.	Keep the boost off while EN_BOOST is inactive; the 1k limits current from the GPIO.
7	R6, R7, C7, C8	82k FREQ resistor, 10k/22 nF COMP network, 47 nF soft-start.	Set the switching frequency, loop compensation and soft-start.
8	C9, C10	Output capacitors 22 uF/35 V and 10 uF/50 V.	Boost output filter.

Figure 6 - Battery boost to 5 V (U41, TPS61088)

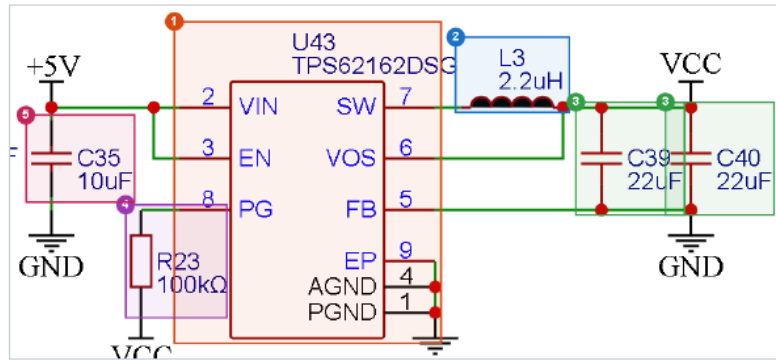
Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-06	1.0	2026-07-19	Battery boost to 5 V (U41, TPS61088)	6 of 26

Block function: Boosts the battery to 5VBOOST. **Requirement covered:** 5 V rail

No	Component	What it is	What it does
1	U41	Boost converter TPS61088 (VIN = VBAT, VOUT = 5VBOOST, EN = ENA).	Generates 5 V from the battery. Active when the ENA signal from the power-latch circuit is high.
2	L2	2.2 uH inductor.	Boost inductor between VBAT and the SW node.
3	R90, R89	FB divider 189k / 56k.	Sets the output to about $1.204 \text{ V} \times (1 + 189/56) = 5.3 \text{ V}$ (calculated from the resistor values).
4	R34	255k resistor on FSW.	Sets the switching frequency.
5	R30	100k resistor on ILIM.	Sets the peak current limit.
6	C74, C75, C31	Internal VCC capacitor, soft-start capacitor and bootstrap capacitor.	Support parts for the IC.
7	R40, C76, C77	COMP compensation network.	Stabilizes the boost control loop.
8	C64, C67, C68, C71, C36, C37	VBAT input and 5VBOOST output capacitors.	Decoupling on the input and output sides of the boost.

3.3 V rail

Figure 7 - 5 V to 3.3 V buck (U43)

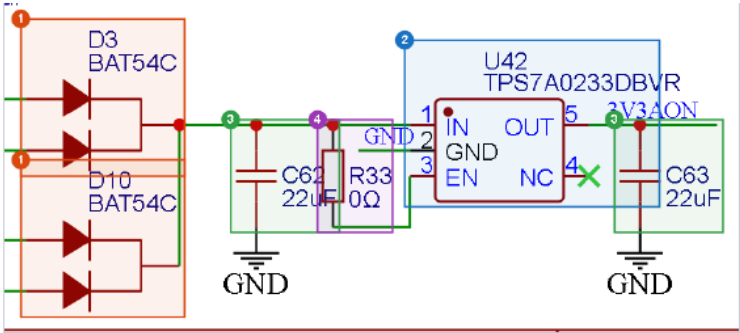


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-07	1.0	2026-07-19	5 V to 3.3 V buck (U43)	7 of 26

Block function: Buck that generates the 3.3 V VCC rail. **Requirement covered:** 3.3 V rail

No	Component	What it is	What it does
1	U43	Buck TPS62162 (VIN = +5V, EN = +5V, VOS = VCC).	Steps +5V down to 3.3 V (fixed-voltage version, FB to GND). Always on because EN is tied to +5V.
2	L3	2.2 uH inductor.	Buck inductor from the SW node to VCC.
3	C39, C40	10 uF / 22 uF capacitors on VCC.	Bulk capacitors of the 3.3 V output.
4	R23	100k pull-up on PG.	PG is not connected to any other component in this netlist (pull-up only).
5	C35	+5V capacitor, PG pull-up and VCC output capacitors.	Decoupling of the input/output of the 3.3 V buck; R23 is the pull-up on the PG pin.

Figure 8 - Always-on 3.3 V LDO and its source selection (U42)

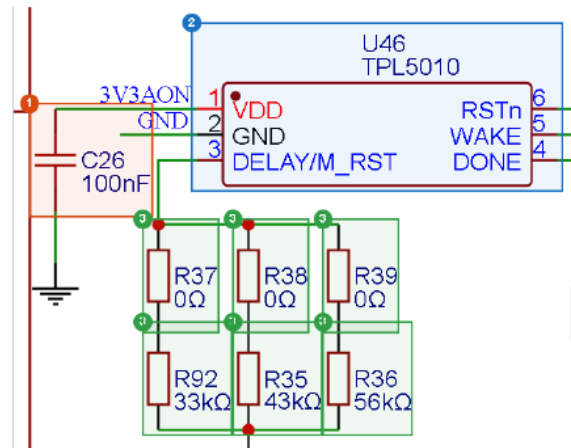


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-08	1.0	2026-07-19	Always-on 3.3 V LDO and its source selection (U42)	8 of 26

Block function: The 3V3AON LDO and its source selection. **Requirement covered:** 3.3 V rail

No	Component	What it is	What it does
1	D3, D10	Dual diode BAT54C (common cathode).	D3 ORs VBAT_IN and 5VBUCK; D10 ORs 5VEXT and VBUS (USB). The result is the net that feeds the LDO input.
2	U42	LDO TPS7A0233 (3.3 V, very low quiescent current).	Converts the available source into 3V3AON. EN is pulled to the input through R33 (0 ohm), so it is always enabled.
3	C62, C63	22 uF and 100 nF input/output capacitors.	Decoupling of the LDO and the 3V3AON rail.
4	R33	0 ohm resistor between the input and the EN pin of U42.	Ties EN to the input so the LDO is always enabled (can be removed if a separate EN is needed).

Figure 9 - TPL5010 timer/watchdog (U46)

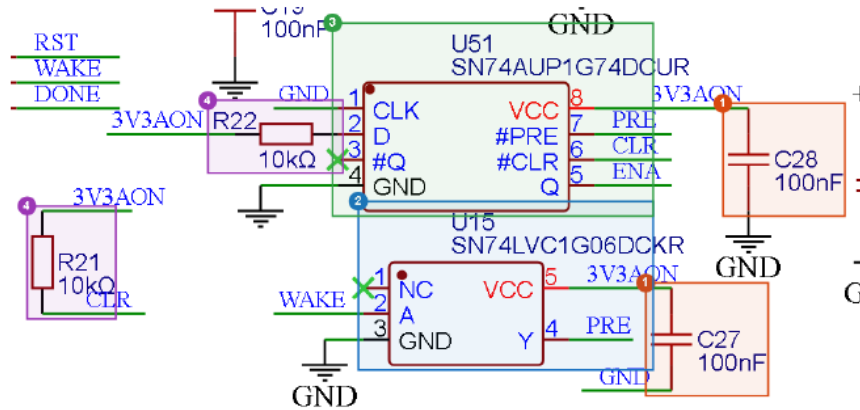


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-09	1.0	2026-07-19	TPL5010 timer/watchdog (U46)	9 of 26

Block function: Nano-power timer/watchdog and its interval selector. **Requirement covered:** 3.3 V rail / power latch

No	Component	What it is	What it does
1	C26	22 uF and 100 nF input/output capacitors.	Decoupling of the LDO and the 3V3AON rail.
2	U46	Nano-power timer TPL5010.	Generates periodic WAKE pulses and works as a watchdog: DONE from the ESP32 (IO17) signals that the firmware is still alive, RSTn goes to RST (ESP32 reset).
3	R35, R36, R92, R37, R38, R39	43k / 56k / 33k resistors to GND with three 0 ohm jumpers.	Select the TPL5010 timer interval through the DELAY/M_RST pin (one 0 ohm jumper is fitted for the chosen interval).

Figure 10 - ENAlatch (U51) and open-drain inverter (U15)

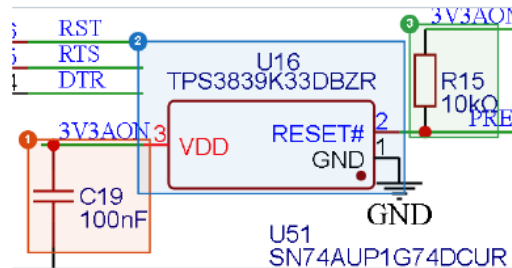


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-10	1.0	2026-07-19	ENAlatch (U51) and open-drain inverter (U15)	10 of 26

Block function: ENAlatch flip-flop and open-drain inverter. **Requirement covered:** 3.3 V rail / power latch

No	Component	What it is	What it does
1	C27, C28	22 uF and 100 nF input/output capacitors.	Decoupling of the LDO and the 3V3AON rail.
2	U15	Open-drain inverter SN74LVC1G06.	Turns WAKE from U46 into a pull-down on net PRE (R15 pull-up to 3V3AON).
3	U51	D flip-flop SN74AUP1G74 (CLK to GND, Q = ENA).	Used as a latch: PRE low sets ENA, CLR low from the ESP32 (IO16) clears ENA. ENA controls U14 (load switch) and U41 (5 V boost).
4	R21, R22	10k pull-ups to 3V3AON on PRE, CLR and the flip-flop D input.	Keep the default logic levels.

Figure 11 - 3V3AON voltage supervisor (U16)



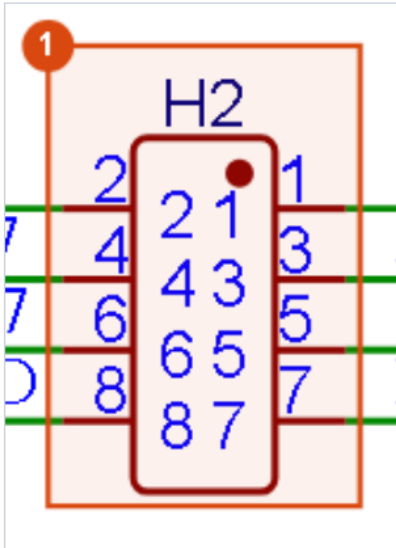
Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-11	1.0	2026-07-19	3V3AON voltage supervisor (U16)	11 of 26

Block function: Voltage supervisor of 3V3AON. **Requirement covered:** 3.3 V rail / power latch

No	Component	What it is	What it does
1	C19	22 uF and 100 nF input/output capacitors.	Decoupling of the LDO and the 3V3AON rail.
2	U16	Voltage supervisor TPS3839K33 (threshold variant for a 3.3 V rail).	RESET# output is connected to net PRE; supervises 3V3AON.
3	R15	10k pull-ups to 3V3AON on PRE, CLR and the flip-flop D input.	Keep the default logic levels.

MQ heater supply

Figure 12 - Sensor module connector H2 (one connector as example)

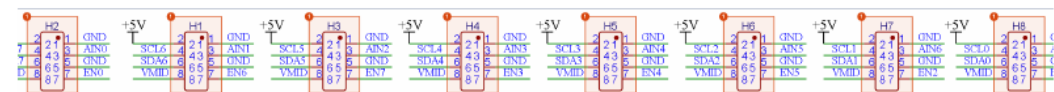


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-12	1.0	2026-07-19	Sensor module connector H2 (one connector as example)	12 of 26

Block function: One sensor module connector as an example of every pin function. **Requirement covered:** MQ heater supply and MQ sensor interface

No	Component	What it is	What it does
1	H2	Sensor module connectors (H1..H8).	Each connector carries: pin 2 = +5V (module/heater supply), pin 8 = VMID (bias), pin 3 = AINx (analog signal to the ADC), pins 4/6 = SCLx/SDAx (I2C channel from the mux), pin 7 = ENx (module enable), pins 1/5 = GND. The heater switch and sensor load resistor are not on this board (assumed to be on the sensor module; to be confirmed).

Figure 13 - Sensor module connectors H1-H8

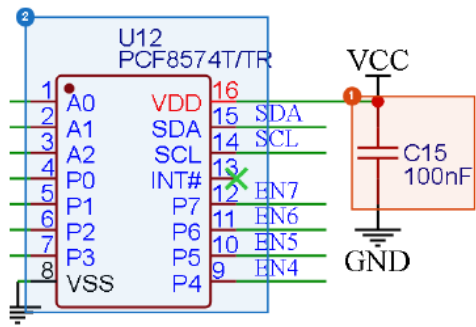


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-13	1.0	2026-07-19	Sensor module connectors H1-H8	13 of 26

Block function: The eight sensor module connectors H1-H8. **Requirement covered:** MQ heater supply and MQ sensor interface

No	Component	What it is	What it does
1	H1, H2, H3, H4, H5, H6, H7, H8	Sensor module connectors (H1..H8).	Each connector carries: pin 2 = +5V (module/heater supply), pin 8 = VMID (bias), pin 3 = AINx (analog signal to the ADC), pins 4/6 = SCLx/SDAx (I2C channel from the mux), pin 7 = ENx (module enable), pins 1/5 = GND. The heater switch and sensor load resistor are not on this board (assumed to be on the sensor module; to be confirmed).

Figure 14 - Sensor module enable I/O expander (U12)



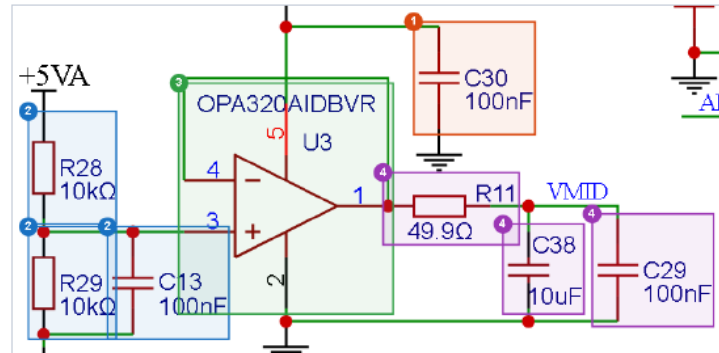
Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-14	1.0	2026-07-19	Sensor module enable I/O expander (U12)	14 of 26

Block function: I/O expander that enables the sensor modules. **Requirement covered:** MQ heater supply (module enable)

No	Component	What it is	What it does
1	C15	100 nF capacitors on VCC.	Decoupling close to each 3.3 V IC.
2	U12	I/O expander PCF8574.	Outputs P0-P7 become EN0-EN7 to enable the sensor modules one by one.

MQ sensor bridge / analog circuitry

Figure 15 - VMID bias of about 2.5 V (U3)

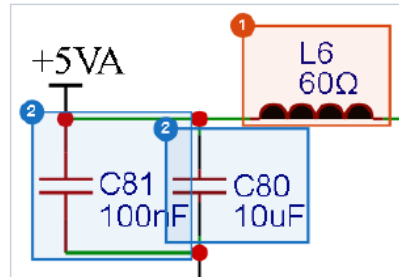


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-15	1.0	2026-07-19	VMID bias of about 2.5 V (U3)	15 of 26

Block function: Generator of the VMID bias of about 2.5 V. **Requirement covered:** MQ sensor bridge / analog circuitry

No	Component	What it is	What it does
1	C30	10 uF and 100 nF capacitors on +5VA.	Decoupling for the ADC, op-amps and reference.
2	R28, R29, C13	10k / 10k divider and 100 nF.	Creates half of +5VA (about 2.5 V).
3	U3	Op-amp OPA320 as a buffer.	Buffers the mid-voltage into a low-impedance VMID source.
4	R11, C29, C38	49.9 ohm resistor and capacitors.	Isolate the buffer output and keep it stable with capacitive loads; VMID goes to pin 8 of every sensor connector.

Figure 16 - Analog +5VAsupply filter (L6)

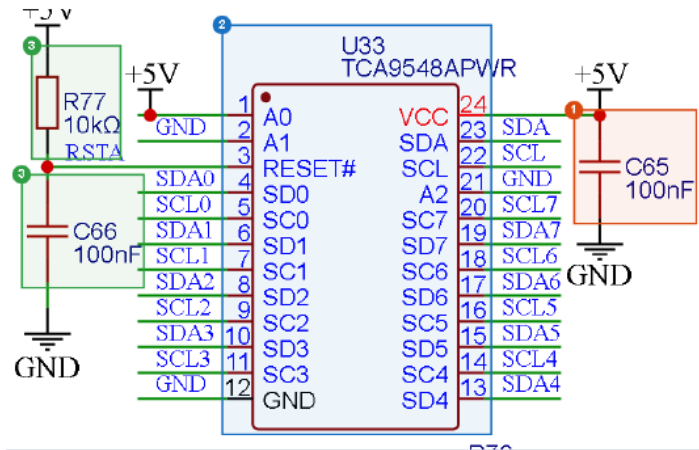


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-16	1.0	2026-07-19	Analog +5VAsupply filter (L6)	16 of 26

Block function: Analog +5VAsupply filter. **Requirement covered:** MQ sensor bridge / analog circuitry

No	Component	What it is	What it does
1	L6	60 ohm ferrite bead.	Separates digital noise from the analog +5VA rail.
2	C80, C81	10 uF and 100 nF capacitors on +5VA.	Decoupling for the ADC, op-amps and reference.

Figure 17 - Sensor module I2C multiplexer (U33)

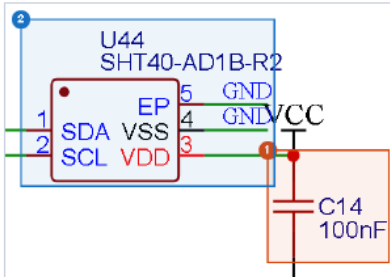


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-17	1.0	2026-07-19	Sensor module I2C multiplexer (U33)	17 of 26

Block function: I2C multiplexer for the eight sensor modules. **Requirement covered:** MQ sensor bridge / analog circuitry

No	Component	What it is	What it does
1	C65	10 uF and 100 nF capacitors on +5V.	Decoupling of the +5V rail.
2	U33	I2C multiplexer TCA9548A(8 channels).	Splits the main I2C bus into eight channels SDA0-7/SCL0-7 toward the modules; RESET# is held high by R77.
3	R77, C66	10k pull-up and 100 nF on RSTA.	Keep the mux from being reset unless intended.

Figure 18 - Temperature and humidity sensor (U44)



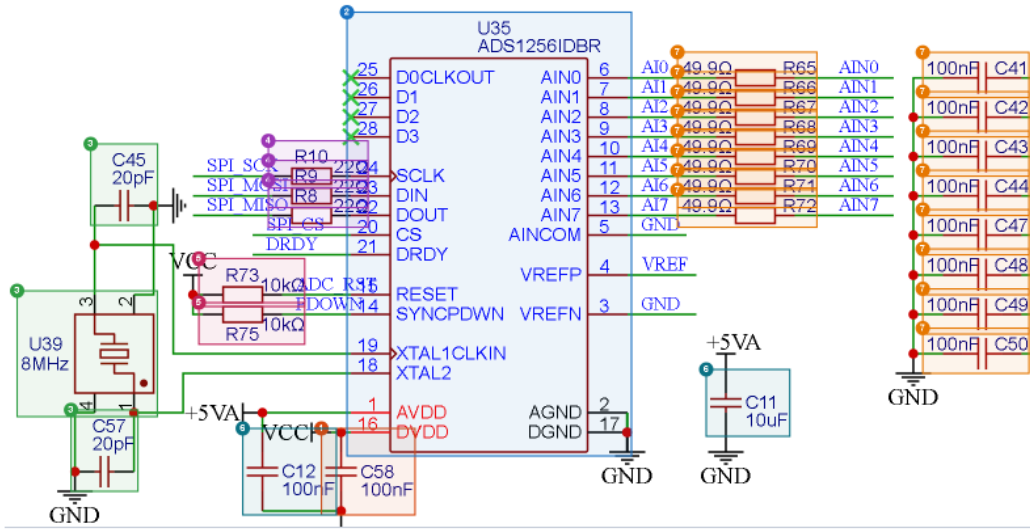
Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-18	1.0	2026-07-19	Temperature and humidity sensor (U44)	18 of 26

Block function: Environment sensor inside the board. **Requirement covered:** MQ sensor bridge / analog circuitry (environment compensation)

No	Component	What it is	What it does
1	C14	100 nF capacitors on VCC.	Decoupling close to each 3.3 V IC.
2	U44	Temperature/humidity sensor SHT40.	Measures temperature and humidity inside the board on the main I2C bus.

ADC

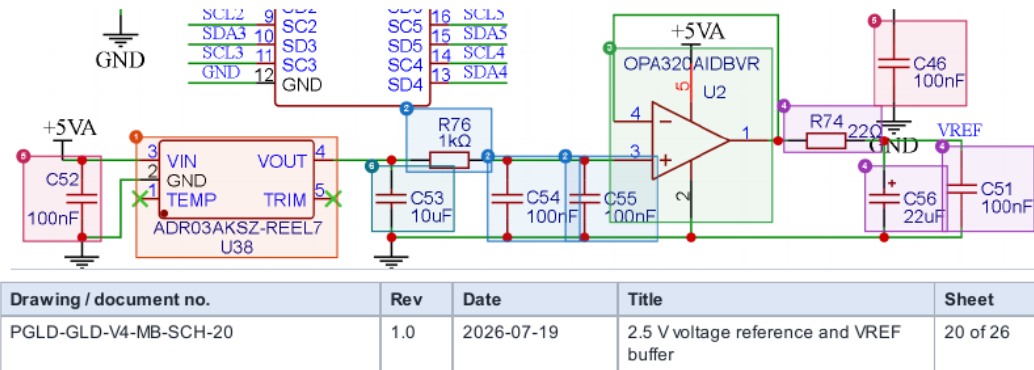
Figure 19 - ADS1256 ADC with input filters (AIN), clock and decoupling



Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-19	1.0	2026-07-19	ADS1256 ADC with input filters (AIN), clock and decoupling	19 of 26

Block function: ADS1256 with its input filters, clock and decoupling. **Requirement covered:** ADC; MQ sensor bridge / analog circuitry

No	Component	What it is	What it does
1	C58	100 nF capacitors on VCC.	Decoupling close to each 3.3 V IC.
2	U35	Delta-sigma ADC ADS1256 (8 channels, 24 bit).	Converts the AI0-AI7 voltages to digital data. AVDD = +5VA, DVDD = VCC, VREFP = VREF, VREFN and AINCOM = GND. Controlled over SPI (CS, SCLK, DIN, DOUT) + DRDY.
3	U39, C45, C57	4-pin SMD crystal with two 20 pF capacitors.	ADC clock source (8 MHz according to the U39 label on the schematic).
4	R8, R9, R10	22 ohm series resistors on MISO, MOSI, SCK.	Damp ringing/EMI on the SPI lines to the ADC.
5	R73, R75	10k pull-ups on RESET and SYNC/PDWN.	Keep the ADC active by default; the ESP32 can pull them low (ADC_RST, PDOWN).
6	C11, C12	10 uF and 100 nF capacitors on +5VA.	Decoupling for the ADC, op-amps and reference.
7	R65, R66, R67, R68, R69, R70, R71, R72, C41, C42, C43, C44, C47, C48, C49, C50	49.9 ohm series resistor and 100 nF capacitor to GND on every channel.	About 32 kHz low-pass (calculated) that suppresses noise and protects the ADC inputs; AINx from the connector becomes AIx at the ADS1256.

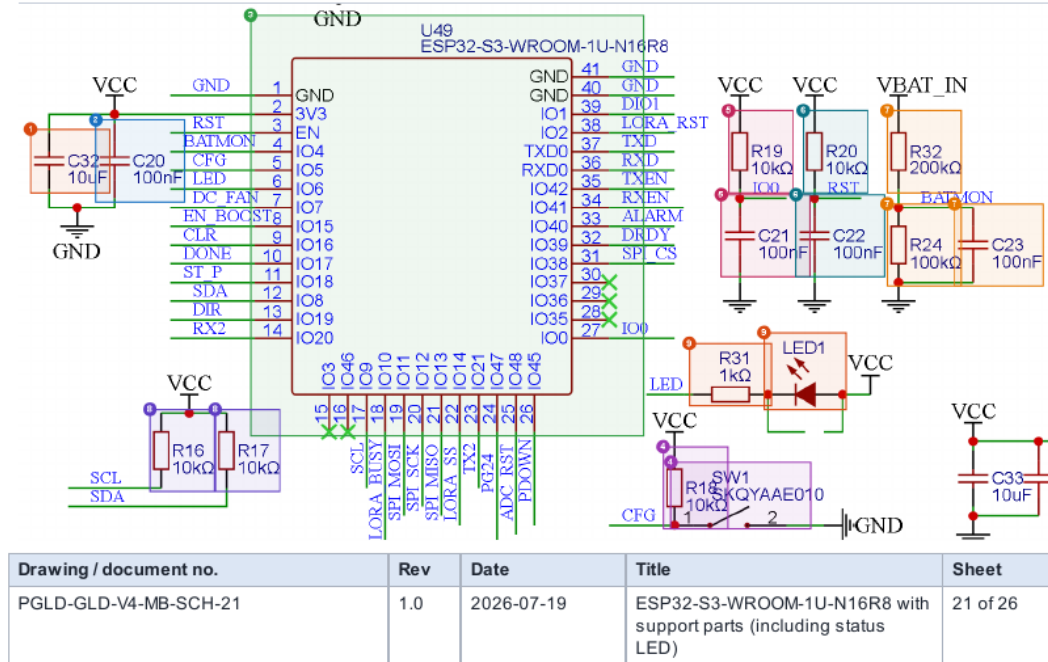
Figure 20 - 2.5 V voltage reference and VREF buffer

Block function: 2.5 V reference and VREF buffer. **Requirement covered:** ADC

No	Component	What it is	What it does
1	U38	Voltage reference ADR03 (2.5 V).	Precision VREF source for the ADC (VIN = +5VA).
2	R76, C54, C55	1k resistor and 2x 100 nF.	Low-pass filter of the reference noise before the buffer.
3	U2	Op-amp OPA320 as a buffer (voltage follower).	Buffers VREF so it is stiff against the load and the switching current of the ADC.
4	R74, C56, C51	22 ohm resistor, 22 uF tantalum, 100 nF.	Isolation and reservoir capacitors on the ADC VREF pin.
5	C46, C52	10 uF and 100 nF capacitors on +5VA.	Decoupling for the ADC, op-amps and reference.
6	C53	10 uF capacitor on VREF_2V5 (output of U38).	Reservoir/filter of the 2.5 V reference output.

ESP32-S3

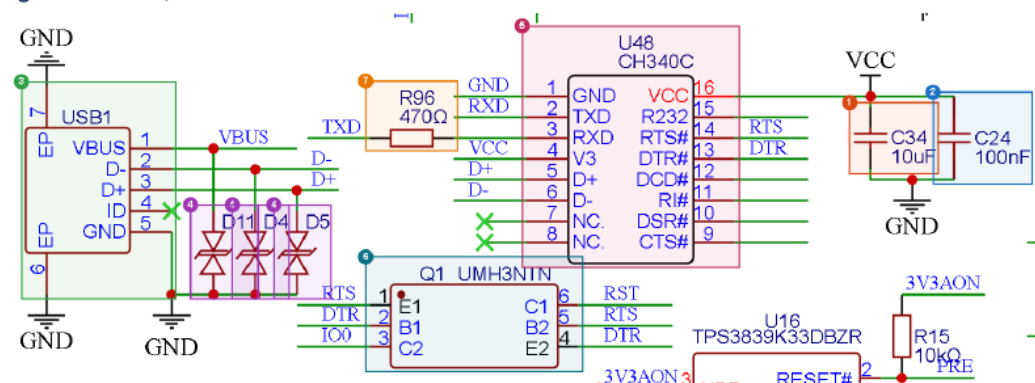
Figure 21 - ESP32-S3-WROOM-1U-N16R8 with support parts (including status LED)



Block function: Main microcontroller with reset, boot, LED and I2C pull-ups. **Requirement covered:** ESP32-S3; LED / buzzer driver (LED)

No	Component	What it is	What it does
1	C32	10 uF / 22 uF capacitors on VCC.	Bulk capacitors of the 3.3 V output.
2	C20	100 nF capacitors on VCC.	Decoupling close to each 3.3 V IC.
3	U49	Module ESP32-S3-WROOM-1U-N16R8 (external antenna, 16 MB flash, 8 MB PSRAM).	The brain of the GLD: reads the ADC (SPI_CS/DRDY/ADC_RST/PDOWN), controls the LoRa radio (LORA_SS/RST/BUSY/DIO1/TXEN/RXEN), I2C (SDA/SCL), RS-485 (TX2/RX2/DIR), alarm, fan, LED and boost, and reads BATMON, PG24, ST_P and CFG.
4	SW1, R18	CFG button with a 10k pull-up.	Configuration button: pulls CFG to GND when pressed.
5	R19, C21	10k pull-up and 100 nF on IO0.	Keep IO0 high (normal boot); Q1 pulls it low in download mode.
6	R20, C22	10k pull-up and 100 nF on EN/RST.	RC reset circuit for the ESP32; also driven by the TPL5010.
7	R32, R24, C23	200k / 100k divider to BATMON with 100 nF.	Scales VBAT_IN down so the ESP32 ADC (IO4) can read it safely; the 1/3 ratio matches the firmware formula 'avg x 3'.
8	R16, R17	10k I2C pull-ups to VCC.	Pull-ups for SDA and SCL.
9	LED1, R31	0603 LED (19-217/BHC-ZL1M2RY/3T) with a 1k resistor.	Lights when the ESP32 pulls the LED pin (IO6) to GND; anode to VCC (active-low).

Figure 22 - USB, CH340C and auto-reset



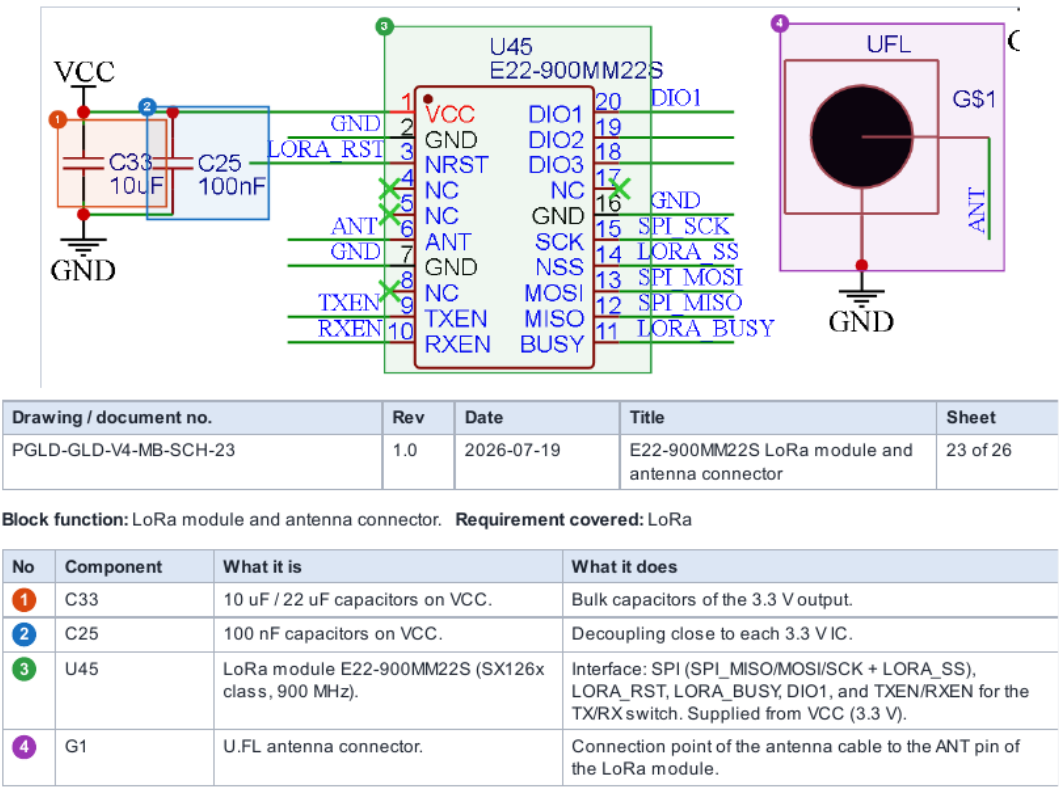
Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-22	1.0	2026-07-19	USB, CH340C and auto-reset	22 of 26

Block function: USB connector, ESD protection, CH340C and auto-reset. **Requirement covered:** ESP32-S3 (programming)

No	Component	What it is	What it does
1	C34	10 uF / 22 uF capacitors on VCC.	Bulk capacitors of the 3.3 V output.
2	C24	100 nF capacitors on VCC.	Decoupling close to each 3.3 V IC.
3	USB1	Micro-USB connector.	Programming and serial port to the PC; VBUS also feeds the 3V3AON source OR-ing (3.3 V rail).
4	D4, D5, D11	ESD protection on D-, D+ and VBUS.	Protects the USB lines from electrostatic discharge.
5	U48	USB-UART bridge CH340C (VCC 3.3 V).	Translates USB to the ESP32 UART0 (TXD/RXD).
6	Q1	Dual NPN with resistors, UMH3N.	Classic auto-reset circuit: DTR/RTS from the CH340 drive IO0 and RST of the ESP32 during upload.
7	R96	470 ohm resistor.	Series resistor on the TXD line from the ESP32 to RXD of the CH340C.

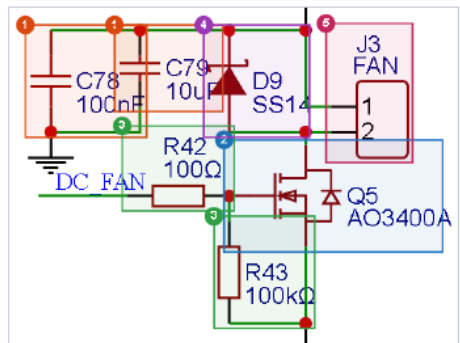
LoRa

Figure 23 - E22-900MM22S LoRa module and antenna connector



Fan driver

Figure 24 - 5 V fan driver (Q5, J3)



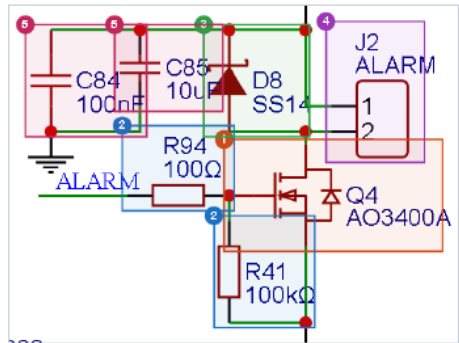
Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-24	1.0	2026-07-19	5 V fan driver (Q5, J3)	24 of 26

Block function: 5 V fan driver. Requirement covered: Fan driver

No	Component	What it is	What it does
1	C79, C78	10 uF and 100 nF capacitors on +5V.	Decoupling of the +5V rail.
2	Q5	N-MOSFET AO3400A as a low-side switch.	Turns on the fan at J3 when DC_FAN is high.
3	R42, R43	100 ohm gate resistor and 100k pull-down.	Gate current limiter and default-off holder.
4	D9	SS14 flyback diode.	Clamps the inductive spike of the fan to +5V.
5	J3	Fan connector (pin 1 = +5V, pin 2 = Q5 drain).	Connection point of the 5 V fan.

LED / buzzer driver

Figure 25 - 24 V alarm output driver (Q4, J2)



Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-25	1.0	2026-07-19	24 V alarm output driver (Q4, J2)	25 of 26

Block function: 24 V alarm output driver. Requirement covered: LED / buzzer driver

No	Component	What it is	What it does
1	Q4	N-MOSFET AO3400A as a low-side switch.	Turns on the load at J2 when ALARM is high; its source goes to GND.
2	R94, R41	100 ohm gate resistor and 100k pull-down.	R94 limits the gate charging current; R41 keeps Q4 off while the GPIO is not yet active.
3	D8	SS14 flyback diode.	Clamps the inductive spike of the load at J2 to +24V.
4	J2	Alarm load connector (pin 1 = +24V, pin 2 = Q4 drain).	Connection point of the 24 V alarm load (load type to be confirmed).
5	C84, C85	100 nF and 10 uF capacitors on +24V.	Decoupling of the +24V rail next to the alarm driver.

2.6.a.8 PCB layout

The layout of the GLD main board is presented in ten views generated from the EasyEDA PCB file of the same project. Each view highlights one aspect required by item 2.6.a.8; numbered markers on the view correspond to the legend printed below it.

Board data

Parameter	Value	Source / status
Board name	Main Board GLD (MotherBoardGLDVer2)	Layout file
Board outline	Circular, Ø 84.0 mm	Dimension in view 04
Copper layers	Top and bottom (two views of copper)	Views 01, 02
Component side	All SMD parts on top; only through-hole connectors pass to the bottom	View 02
Enclosure fixing holes	4 × Ø 3.2 mm on Ø 64 mm pitch circle, at (0, +32), (+32, 0), (0, -32), (-32, 0) mm from centre	View 05
Additional holes	4 × Ø 3.0 mm near the board edge ($\approx \pm 39$ mm from centre)	View 05 — purpose (casing screw or standoff) to be confirmed against the mechanical drawing
Ground	GND pour on top and bottom; 176 of 400 vias stitch the GND pours	Views 09, 10
Laminate, thickness, copper weight	Not stated in the layout package	To be provided (FR-4 grade, UL 94 rating, Tg, CTI) — see 6.b / 6.c
Layout revision / date	Not stated in the layout package	To be added to the drawing table

Drawing register

Drawing no.	Title	Requirement covered	Rev / date
PGLD-GLD-V4-MB-PCB-01	Top layer	Top layer	TBC
PGLD-GLD-V4-MB-PCB-02	Bottom layer	Bottom layer	TBC
PGLD-GLD-V4-MB-PCB-03	Component placement	Component placement	TBC
PGLD-GLD-V4-MB-PCB-04	Board dimensions	Board dimensions	TBC
PGLD-GLD-V4-MB-PCB-05	Mounting holes	Mounting holes	TBC
PGLD-GLD-V4-MB-PCB-06	Connectors	Connectors	TBC
PGLD-GLD-V4-MB-PCB-07	High-current / power area	High-current / power area	TBC
PGLD-GLD-V4-MB-PCB-08	Sensor interface	Sensor interface	TBC
PGLD-GLD-V4-MB-PCB-09	Grounding arrangement — top	Grounding arrangement	TBC
PGLD-GLD-V4-MB-PCB-10	Grounding arrangement — bottom	Grounding arrangement	TBC

Coverage of the required layout content

Required content (2.6.a.8)	Where shown	Status
Top layer	View 01	Provided
Bottom layer	View 02	Provided
Component placement	View 03	Provided
Board dimensions	View 04	Provided
Mounting holes	View 05	Provided — purpose of 4 × Ø 3.0 mm holes to confirm

Required content (2.6.a.8)	Where shown	Status
Connectors	View 06	Provided
High-current / power area	View 07	Provided
Sensor interface	View 08	Provided
Grounding arrangement	Views 09, 10	Provided
One set per board (Main, Sensor, Alarm, Terminal)	Main board only	Open — sensor-module board layout to be added

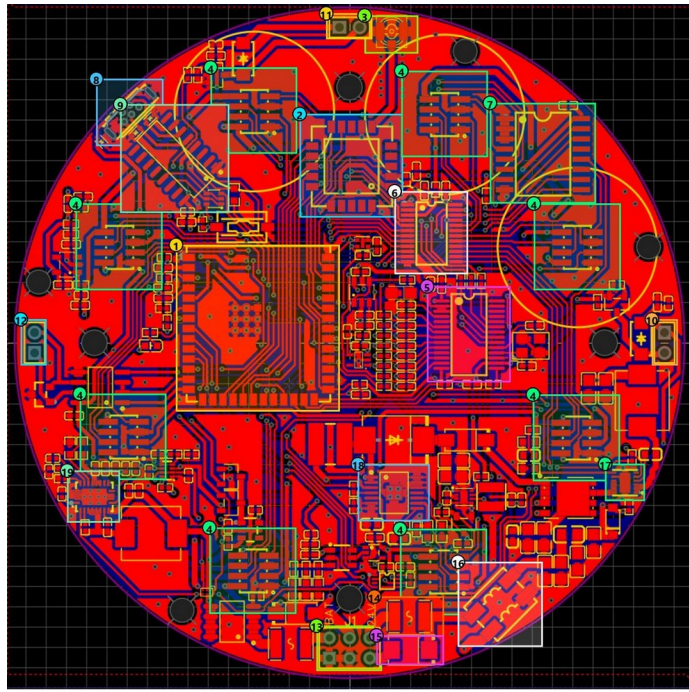


Figure 8-1. Top layer — Main Board GLD

Drawing no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-PCB-01	TBC	TBC	Top layer	1 of 10

No	Item	Description
1	ESP32-S3	Main MCU module: runs the GLD firmware; SPI to LoRa and ADC, I ² C to sensors, GPIO for alarm/fan.
2	LoRa E22-900MM22S	LoRa radio for communication with the cluster head.
3	U.FL antenna connector	Connection to the external LoRa antenna.
4	Sensor ports (8)	Eight 2 × 4 headers around the board, one per sensor module (yellow circle = module outline).
5	ADS1256	24-bit ADC reading the eight analog sensor channels.
6	TCA9548A	8-channel I ² C multiplexer, one channel per sensor port.
7	PCF8574	I/O expander providing the enable signals EN0–EN7 for each sensor.
8	Micro-USB	USB port for firmware flashing and serial debug.
9	CH340C	USB-to-UART converter for programming and serial monitor.
10	ALARM connector	24 V siren/beacon output.
11	FAN connector	5 V fan output.
12	RS-485 connector	RS-485 bus (A/B).
13	VIN connector	Power input: 24V+, 24V–, 5VEXT, GND, BAT (battery).
14	PTC fuse	Limits current on the 24 V line under short circuit.
15	TVS	Clamps voltage surges on the 24 V input.
16	Common-mode filter	Filters common-mode noise on the 24 V input.
17	Buck 24 V → 5 V	LMR51450 regulator stepping 24 V down to 5 V.
18	Battery boost → 24 V	TPS61175 boosting the battery to the +24V rail.
19	Battery boost → 5 V	TPS61088 generating 5 V from the battery.

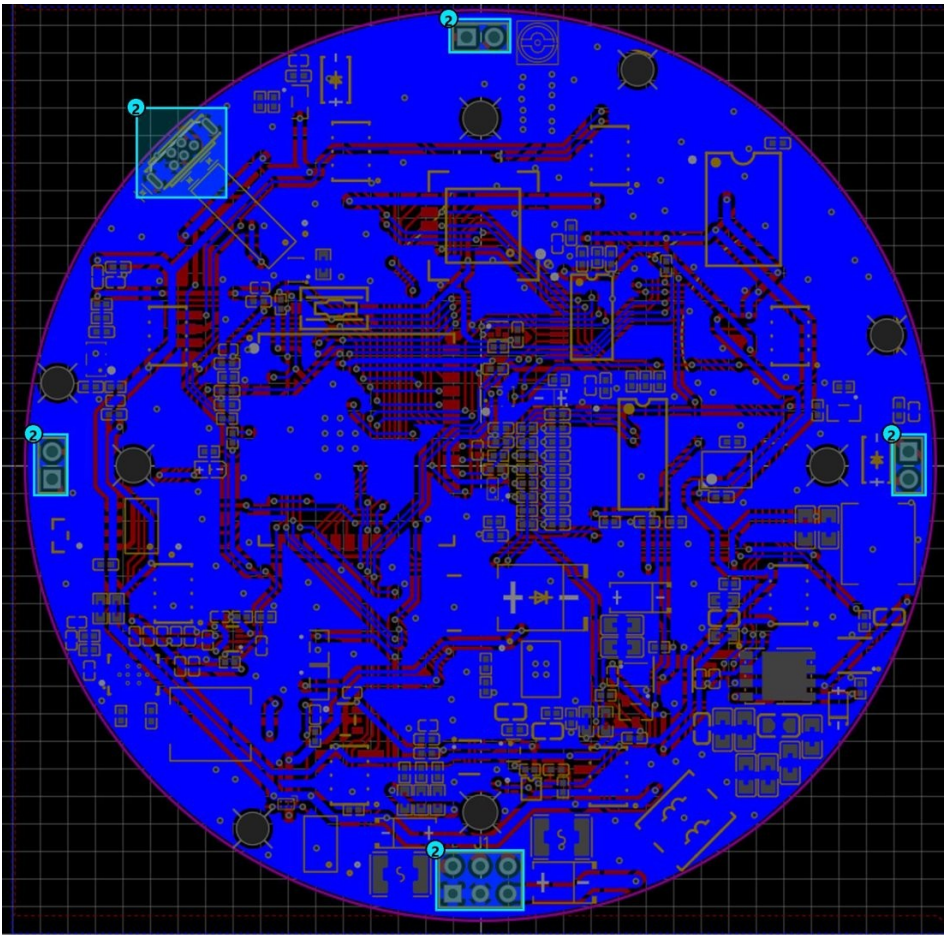


Figure 8-2. Bottom layer — Main Board GLD

Drawing no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-PCB-02	TBC	TBC	Bottom layer	2 of 10

No	Item	Description
1	Bottom GND pour (solid blue area)	GND plane filling almost the whole bottom side; connected to the top pour through vias.
2	Through-hole connectors	The only parts passing through to the bottom; all SMD parts are on the top side.

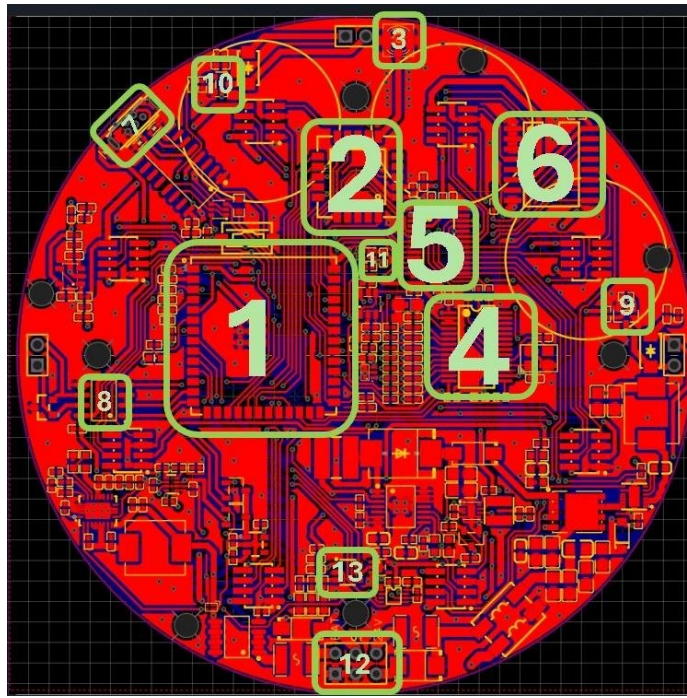


Figure 8-3. Component placement — Main Board GLD

Drawing no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-PCB-03	TBC	TBC	Component placement	3 of 10

No	Item	Description
1	ESP32-S3	Main MCU: runs the firmware and controls all blocks.
2	LoRa (E22-900MM22S)	LoRa radio for communication with the cluster head.
3	Antenna (U.FL)	External LoRa antenna connector.
4	ADS1256	24-bit ADC for the eight analog sensor channels.
5	TCA9548	8-channel I ² C mux, one channel per sensor port.
6	PCF8574	I/O expander for the enable signals EN0–EN7.
7	USB	Micro-USB + CH340C for firmware flashing and serial monitor.
8	RS-485	THVD1410 transceiver for the RS-485 bus.
9	Alarm driver	MOSFET switching the 24 V siren/beacon.
10	Fan driver	MOSFET switching the 5 V fan.
11	SHT40	Temperature and humidity sensor inside the enclosure.
12	Power block	Power input and protection (VIN connector, fuse, TVS, filter).
13	TPL5010	Watchdog timer that wakes the system periodically.

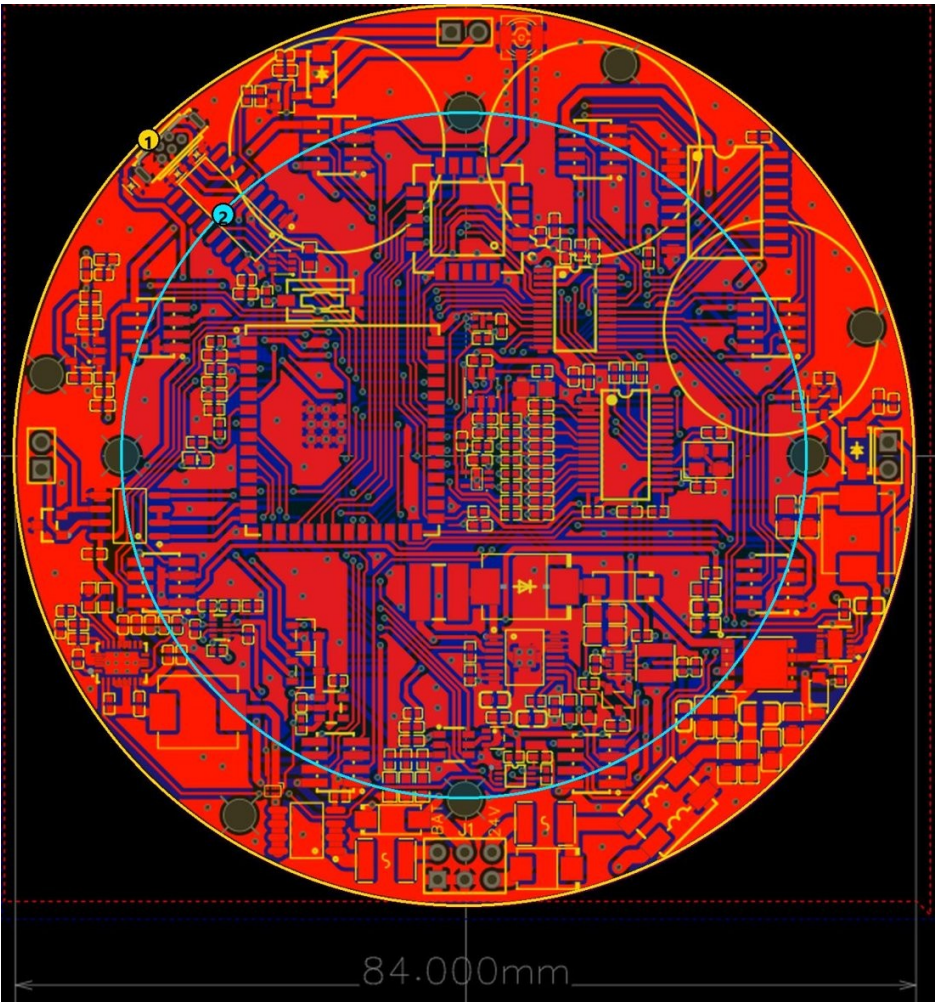


Figure 8-4. Board dimensions — Main Board GLD

Drawing no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-PCB-04	TBC	TBC	Board dimensions	4 of 10

No	Item	Description
1	Board diameter: 84 mm	Circular board (outline) matching the round aluminium enclosure.
2	Enclosure-hole pitch circle: Ø 64 mm	The four enclosure fixing holes lie on this circle, 90° apart.

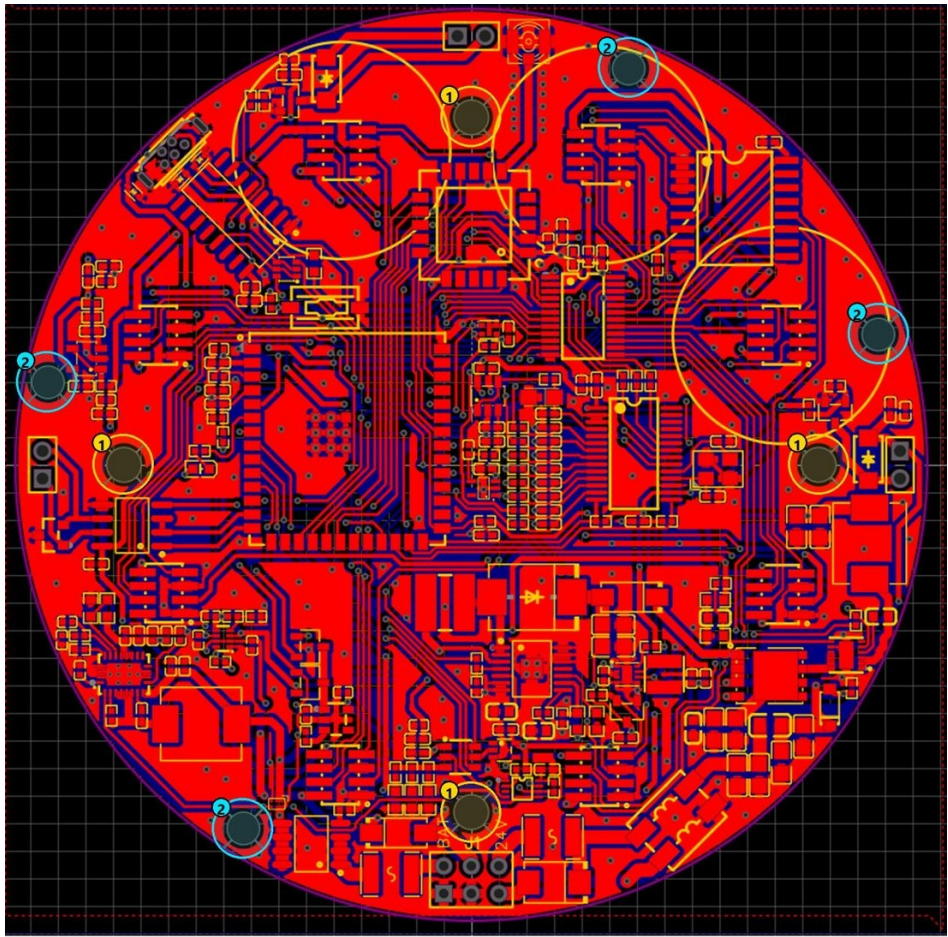


Figure 8-5. Mounting holes — Main Board GLD

Drawing no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-PCB-05	TBC	TBC	Mounting holes	5 of 10

No	Item	Description
1	Enclosure holes (4 × Ø 3.2 mm)	Hole pattern of the enclosure footprint on the Ø 64 mm circle; board-to-enclosure fixing screws. Positions from centre: (0, +32), (+32, 0), (0, -32), (-32, 0) mm.
2	Additional holes (4 × Ø 3.0 mm)	Four further fixing holes near the board edge (≈ ±39 mm from centre). Exact use (enclosure screw or standoff) to be confirmed with the mechanical drawing.

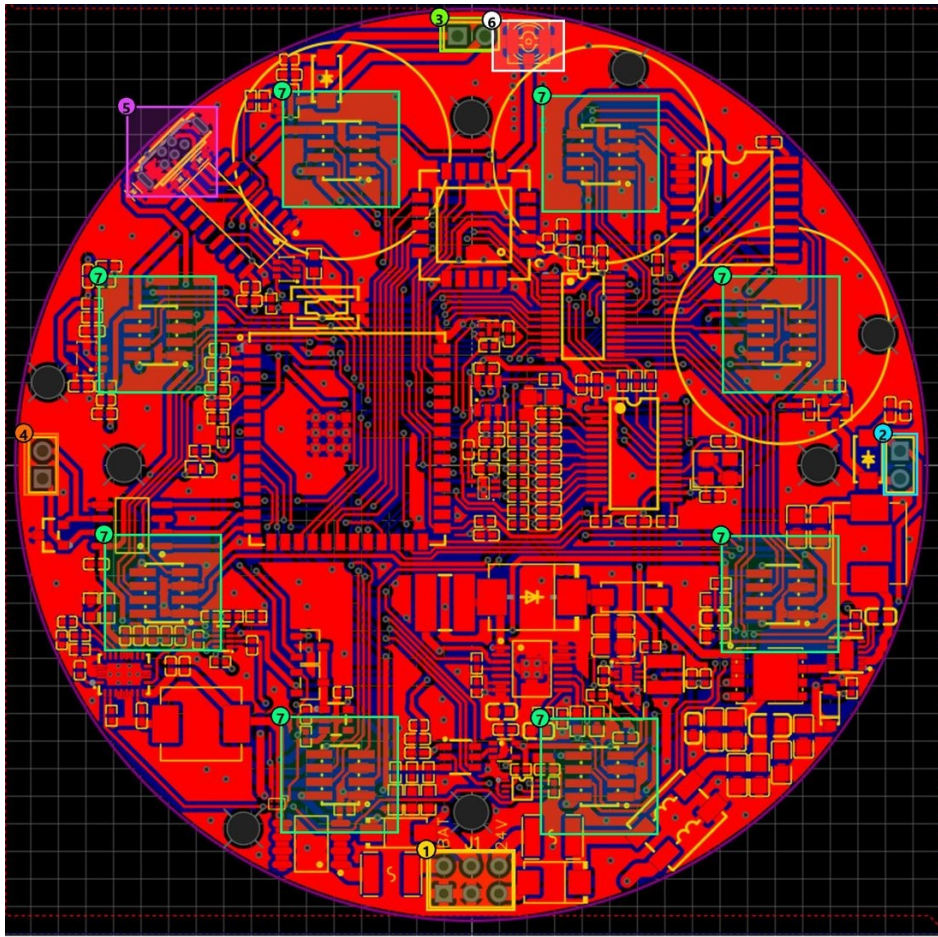


Figure 8-6. Connectors — Main Board GLD

Drawing no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-PCB-06	TBC	TBC	Connectors	6 of 10

No	Item	Description
1	VIN	Power input, 2 × 3 pins: 24V+, 24V-, 5VEXT, GND, BAT (battery).
2	ALARM	24 V siren/beacon output (pin 1 = +24V, pin 2 = MOSFET switch side).
3	FAN	5 V fan output (pin 1 = +5V, pin 2 = MOSFET switch side).
4	RS-485	RS-485 bus: B (pin 1) and A (pin 2).
5	Micro-USB	Firmware flashing and serial debug.
6	U.FL (antenna)	External LoRa antenna connector.
7	Sensor ports (8)	Eight 2 × 4 headers (1.27 mm pitch): GND, +5V, AINx, SCLx, SDAX, ENx, VMID — one sensor module each.

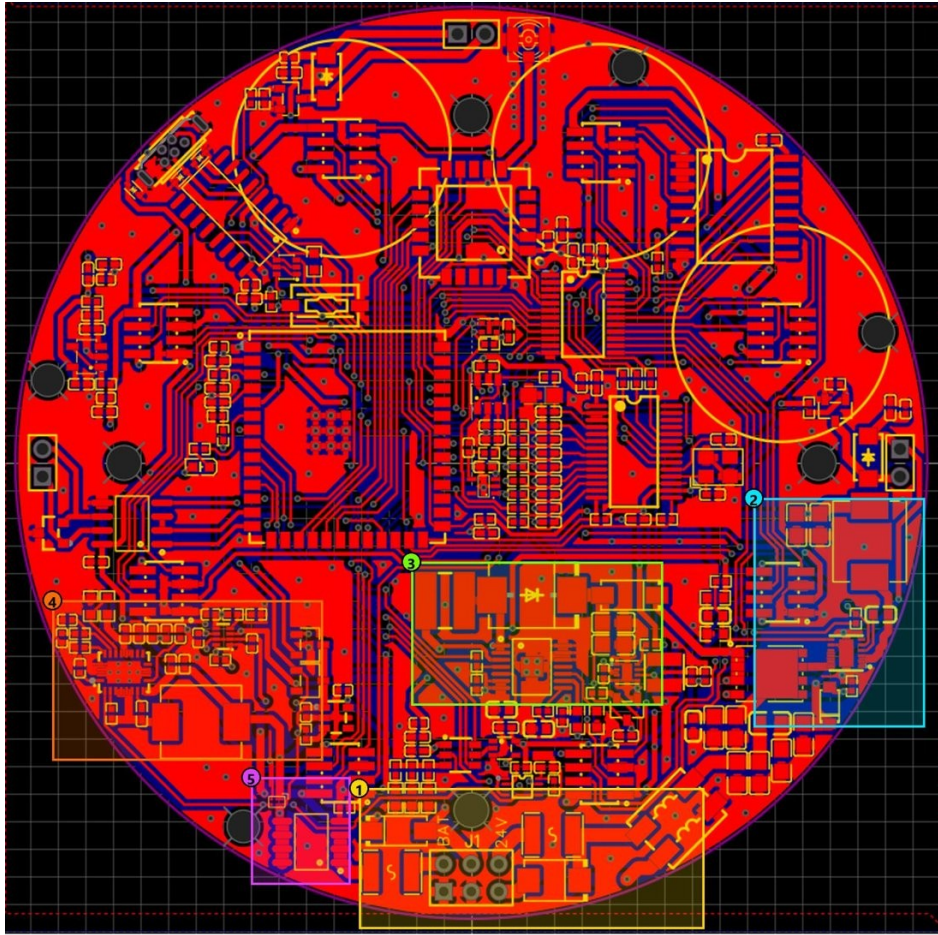


Figure 8-7. High-current / power area — Main Board GLD

Drawing no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-PCB-07	TBC	TBC	High-current / power area	7 of 10

No	Item	Description
1	Power input & protection	VIN connector, fuse, TVS, common-mode filter, ferrite: 24 V entry path.
2	24 V → 5 V converter	LMR51450 buck, 4.7 μ H inductor, P-MOSFET switch.
3	Battery → 24 V converter	TPS61175 boost, 10 μ H inductor, SK36 diode and OR-ing diode to the +24V rail.
4	Battery → 5 V converter + power mux	TPS61088 boost, 2.2 μ H inductor, TPS2116 power mux selecting the 5 V source.
5	Battery protection	Fuse, P-MOSFET, load switch: disconnects the battery when the system sleeps.

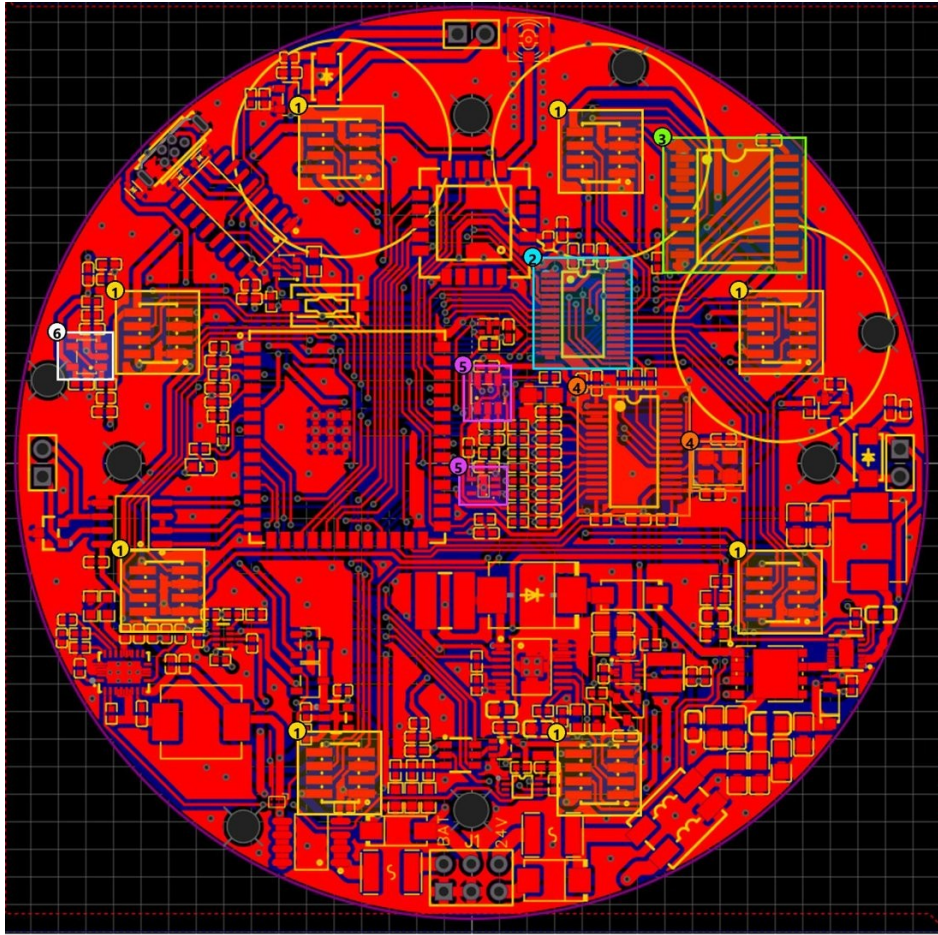


Figure 8-8. Sensor interface — Main Board GLD

Drawing no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-PCB-08	TBC	TBC	Sensor interface	8 of 10

No	Item	Description
1	Sensor ports (8)	Eight headers: +5V, GND, VMID, ENx, SDAX/SCLx (I ² C per channel), AINx (analog to ADC).
2	TCA9548A	I ² C mux splitting the ESP32 I ² C bus into 8 channels for 8 sensors.
3	PCF8574	I/O expander generating EN0-EN7 to switch each sensor module.
4	ADS1256 + 8 MHz crystal	8-channel 24-bit ADC, SPI to ESP32, clocked from an 8 MHz crystal.
5	2.5 V reference + buffer	ADR03 generates VREF, buffered by OPA320 for the ADC.
6	VMID buffer	OPA320 follower generating VMID (analog mid-bias) for the sensor modules.

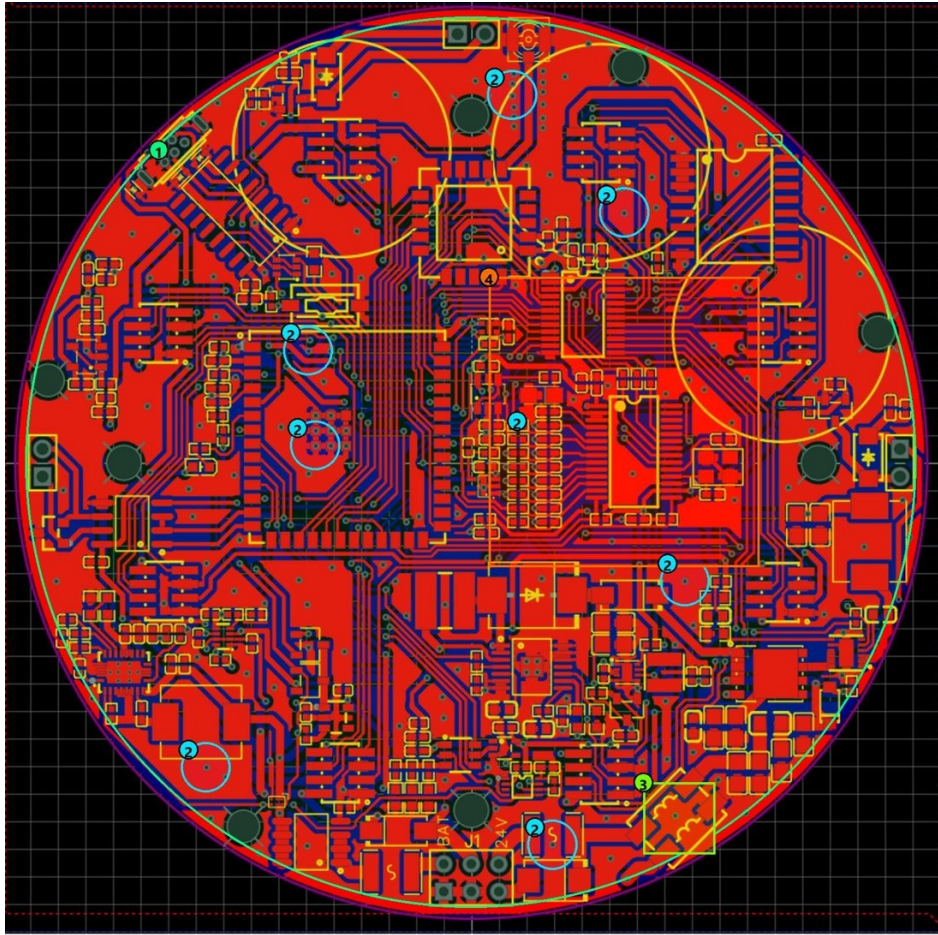


Figure 8-9. Grounding arrangement (top) — Main Board GLD

Drawing no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-PCB-09	TBC	TBC	Grounding arrangement — top	9 of 10

No	Item	Description
1	GND pour	One GND copper area (green line along the board edge) on top and bottom, filling all free area: short return paths and noise shielding.
2	GND vias (examples)	176 of the 400 vias connect the top and bottom GND pours; blue circles mark examples.
3	24V- → GND at the common-mode filter	The 24 V input return (24V-) joins system GND through the return winding of the common-mode choke.
4	Analog area without split ground	Single GND net. The analog area (ADS1256, voltage reference, OPA320) is separated through the +5VA supply (ferrite), not by a split ground.

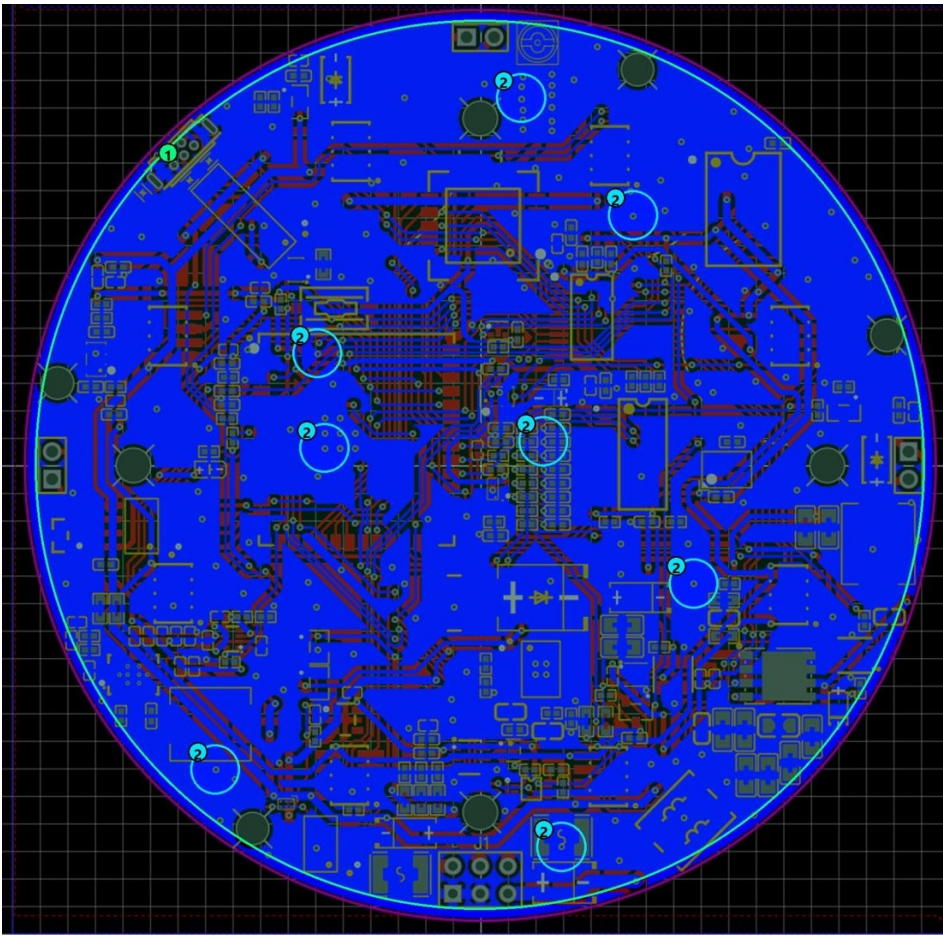


Figure 8-10. Grounding arrangement (bottom) — Main Board GLD

Drawing no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-PCB-10	TBC	TBC	Grounding arrangement — bottom	10 of 10

No	Item	Description
1	GND pour	Bottom GND copper area filling all free area.
2	GND vias (examples)	176 of 400 vias stitch the top and bottom GND pours; blue circles mark examples.

Open items for 2.6.a.8: (a) layout revision and date for the drawing table; (b) PCB laminate grade, UL 94 rating, Tg, CTI, board thickness and copper weight; (c) purpose of the 4 × Ø 3.0 mm holes; (d) layout drawings of the sensor-module board.

2.6.b • Bill of Materials (BOM) for Explosion-Safety-Relevant Components

A complete, itemized electronic-component BOM for both the main board and the external sensor board now exists, exported directly from the EasyEDA/JLPCB source project (manufacturer, manufacturer part number, and LCSC supplier part number for each line item; 89 line items / 214 placed components on the main board, 14 line items / 24 placed components on the sensor board — full tables below). This is real, traceable sourcing data and materially improves on the previous status.

It does not, however, satisfy this checklist item as written. The checklist asks specifically for the explosion-safety-relevant BOM — enclosure, gaskets, terminals, cable entry devices, switches, light sources, battery, potting compound, and plastic parts, each with material grade and Ex/UL/CCC certification status. None of those mechanical/safety items appear in an electronic CAD BOM: the enclosure is present only as a placeholder mechanical symbol (designator U50, no manufacturer or dimensional data attached), and the gas sensor itself (MQ2, designator I1) has no manufacturer or supplier part number recorded — it is sourced outside the LCSC/JLPCB supply chain and its Ex status is unverified. Comparative research on Ex-rated enclosure products from other manufacturers exists internally as a reference for target specifications only — it describes third-party products, not this product's actual components, and is not included here.

DESIGNATOR	VALUE / PART	QTY	MANUFACTURER	MANUFACTURER PART NO.	LCSC #
C1	1uF	1	SAMSUNG	CL21B105KBFNNNE	C28323
C2,C3	100nF	2	YAGEO	CC0805KRX7R9BB104	C49678
C4,C59,C60,C10	10uF	4	muRata	GRM21BR61H106KE43L	C440198
C72,C73	33uF	2	TDK	C2012X5R1A336MTJ00E	C342635
C7	47nF	1	SAMSUNG	CL10B473KB8NNNC	C1622
C8	22nF	1	SAMSUNG	CL10B223KB8NNNC	C21122
C70	33pF	1	SAMSUNG	CL10C330JB8NNNC	C1663
C61,C69	100nF	2	YAGEO	CC0603KRX7R9BB104	C14663
C87,C12,C13,C14,C41,C42,C43,C44,C46,C47,C48,C49,C50,C51,C52,C54,C55,C58,C65,C66,C15,C6,C19,C20,C21,C22,C23,C24,C25,C26,C27,C28,C29,C30,C31,C84,C78,C81	100nF	38	SAMSUNG	CL05B104KO5NNNC	C1525
CN3,CN1	XH-2A-black	2	BOOMELE	XH-2A 黑色	C24193
D1	SMBJ33A	1	RUILON	SMBJ33A	C10219
D2	LBZT52C12T1G	1	LRC	LBZT52C12T1G	C12747
D3,D10	BAT54C	2	CJ	BAT54C	C2135
D4,D5,D11	LES5D5.0CT1G_C5199850	3	UMW	LES5D5.0CT1G	C5199850
D6	SM712	1	DOWO	SM712	C293966
D7,D13	SS54_C22452	2	MDD	SS54	C22452
D12	SK36_C266330	1	GOOD-ARK	SK36	C266330
F1,F2	MINISMDC260F/16	2	Littelfuse	MINISMDC260F/16-2	C16490
G\$1	UFL	1	—	—	—
J1	VIN	1	—	—	C65114
L1	300Ω	1	TDK	ACM7060-301-2PL-TL01	C76580

DESIGNATOR	VALUE / PART	QTY	MANUFACTURER	MANUFACTURER PART NO.	LCSC #
L2	2.2uH	1	SXN	SMDRH105R-2R2NT	C10164
L3	2.2uH	1	APV	ANR3015T2R2M	C6364793
L4	100Ω	1	TDK	MPZ2012S101AT000	C15957
L5	10uH	1	SXN	SMDRH105R-100MT	C9185
LED1	19-217/BHC-ZL1M2RY/3T	1	EVERLIGHT	19-217/BHC-ZL1M2RY/3T	C72041
Q1	UMH3NTN	1	ROHM	UMH3NTN	C123008
Q2	AO4407_C5155210	1	JSMSEMI	AO4407	C5155210
Q3	SI7465DP-T1-GE3	1	VISHAY	SI7465DP-T1-GE3	C499482
R3	187kΩ	1	UNI-ROYAL	0603WAF1873T5E	C22896
R6	82kΩ	1	UNI-ROYAL	0603WAF8202T5E	C23254
R86	19.1kΩ	1	UNI-ROYAL	0603WAF1912T5E	C22897
R87	1kΩ	1	UNI-ROYAL	0603WAF1001T5E	C21190
R89,R36	56kΩ	2	UNI-ROYAL	0603WAF5602T5E	C23206
R90	189k	1	YAGEO	RT0603BRD07189KL	C861174
R92	33kΩ	1	UNI-ROYAL	0603WAF3302T5E	C4216
R96	470Ω	1	UNI-ROYAL	0603WAF4700T5E	C23179
SW1	SKQYAAE010	1	ALPSALPINE	SKQYAAE010	C115352
U1	TPS61175PWP	1	TI	TPS61175PWP	C2879383
U36	LMR51450SDRRR	1	TI	LMR51450SDRRR	C6890985
U40	4.7uH	1	ABC	GSSM06304R7M2AU	C2928306
U41	TPS61088RHLR	1	TI	TPS61088RHLR	C87357
U42	TPS7A0233DBVR	1	TI	TPS7A0233DBVR	C5142805
U43	TPS62162DSG	1	TI	TPS62162DSGR	C40256
U46	TPL5010	1	TI	TPL5010DDCR	C473912
U47	THVD1410DR	1	TI	THVD1410DR	C2671345
U48	CH340C	1	WCH	CH340C	C84681
U50	casingAlGasLeak	1	—	—	—
U51	SN74AUP1G74DCUR	1	TI	SN74AUP1G74DCUR	C140267
USB1	U-D-M5DD-W-1	1	韩国韩荣	U-D-M5DD-W-1	C145795
C11,C53,C32,C33,C34,C35,C36,C37,C16,C17,C18,C38,C85,C79,C80	10uF	15	SAMSUNG	CL05A106MQ5NUNC	C15525
C45,C57	20pF	2	FH	0402CG200J500NT	C1554
C56	22uF	1	Kyocera AVX	TAJA226M010RNJ	C7183
R8,R9,R10,R74	22Ω	4	UNI-ROYAL	0402WGF220JTCE	C25092
R11,R65,R66,R67,R68,R69,R70,R71,R72	49.9Ω	9	FH	RC-02K49R9FT	C140147
R73,R75,R77,R2,R15,R16,R17,R18,R19,R20,R21,R22,R7,R28,R29	10kΩ	15	UNI-ROYAL	0402WGF1002TCE	C25744
R76,R5,R31	1kΩ	3	UNI-ROYAL	0402WGF1001TCE	C11702
U2,U3	OPA320AIDBVR	2	TI	OPA320AIDBVR	C92494
U33	TCA9548APWR	1	TI	TCA9548APWR	C130026
U35	ADS1256IDBR	1	TI	ADS1256IDBR	C28186

DESIGNATOR	VALUE / PART	QTY	MANUFACTURER	MANUFACTURER PART NO.	LCSC #
U38	ADR03AKSZ-REEL7	1	ADI	ADR03AKSZ-REEL7	C206683
U39	8MHz	1	JYJE	3TJ480000NYLBC	C2924618
U44	SHT40-AD1B-R2	1	Sensirion	SHT40-AD1B-R2	C2909890
U45	E22-900MM22S	1	EBYTE	E22-900MM22S	C5333985
U49	ESP32-S3-WROOM-1U-N16R8	1	ESPRESSIF	ESP32-S3-WROOM-1U-N16R8	C3013946
U12	PCF8574T/TR	1	HGSEMI	PCF8574T/TR	C2987288
R13	16kΩ	1	UNI-ROYAL	0402WGF1602TCE	C25759
R14	4.99kΩ	1	YAGEO	RC0402FR-074K99L	C137965
U13	TPS2116DRLR	1	TI	TPS2116DRLR	C3235557
U14	TPS22964CYZPR	1	TI	TPS22964CYZPR	C431579
U15	SN74LVC1G06DCKR	1	TI	SN74LVC1G06DCKR	C7828
U16	TPS3839K33DBZR	1	TI	TPS3839K33DBZR	C96333
C9	22uF	1	TDK	C2012X5R1V226MT000E	C338080
R23,R24,R25,R1,R2 6,R27,R4,R30,R12,R 40,R41,R43	100kΩ	12	UNI-ROYAL	0402WGF1003TCE	C25741
R32	200kΩ	1	UNI-ROYAL	0402WGF2003TCE	C25764
C39,C40,C62,C63,C 64,C67,C68,C5	22uF	8	TAIYO YUDEN	JDK105CBJ226MV-F	C385994
C71	1uF	1	SAMSUNG	CL05A105KA5NQNC	C52923
R33,R37,R38,R39	0Ω	4	UNI-ROYAL	0402WGF0000TCE	C17168
C74	2.2uF	1	SAMSUNG	CL05A225MQ5NSNC	C12530
C75,C76	47nF	2	FH	0402B473K500NT	C82219
R34	255kΩ	1	YAGEO	AC0402DR-07255KL	C226702
R35	43kΩ	1	UNI-ROYAL	0603WAF4302T5E	C23172
C77	4.7nF	1	FH	0402B472K500NT	C1538
D8,D9	SS14	2	MDD	SS14	C2480
J2	ALARM	1	—	—	C124375
Q4,Q5	AO3400A	2	AOS	AO3400A	C20917
R94,R42	100Ω	2	UNI-ROYAL	0402WGF1000TCE	C25076
H2,H1,H3,H4,H5,H6 ,H7,H8	X1321WVS-2X04J-C40D53	8	XKB Connection	X1321WVS-2x04J-C40D53	C2881909
L6	60Ω	1	Sunlord	GZ1005D600TF	C23478

External sensor board (gas-sensing front end):

DESIGNATOR	VALUE / PART	QTY	MANUFACTURER	MANUFACTURER PART NO.	LCSC #
H1	X1321FVS-2X04-C43D48	1	XKB Connection	X1321FVS-2x04-C43D48	C2684732
R9	1kΩ	1	UNI-ROYAL	0402WGF1001TCE	C11702
U5	TPS22919DCKR	1	TI	TPS22919DCKR	C2149796
I1	MQ2	1	—	MQ2	—
C7,C8	10uF	2	SAMSUNG	CL05A106MQ5NUNC	C15525
L1	60Ω	1	Sunlord	GZ1005D600TF	C23478
R1	10MΩ	1	UNI-ROYAL	0402WGF1001TCE	C26086
R2,R3	10kΩ	2	UNI-ROYAL	0402WGF1002TCE	C25744
R7,R8	20kΩ	2	UNI-ROYAL	0402WGF2002TCE	C25765

DESIGNATOR	VALUE / PART	QTY	MANUFACTURER	MANUFACTURER PART NO.	LCSC #
C1,C2,C3,C4,C5,C6	100nF	6	SAMSUNG	CL05B104KO5NNNC	C1525
R4,R5,R6	49.9Ω	3	FH	RC-02K49R9FT	C140147
U1	INA333AIDGKR	1	TI	INA333AIDGKR	C19450
U2	LM321DTR(XBLW)	1	XBLW	LM321DTR(XBLW)	C18723612
U3	MCP4725A0T-E/CH	1	MICROCHIP	MCP4725A0T-E/CH	C144198

Status: Partially available. Full electronic-component BOM with real manufacturer/supplier data now exists (source: EasyEDA/JLPCB export, 11 Sep 2026). The explosion-safety-relevant subset the checklist actually asks for — enclosure, gasket, cable entry device, battery, potting compound, and the gas sensor itself, with material grade and Ex/UL/CCC certification for each — remains not yet compiled. These are mechanical/safety parts, not electronic components: the enclosure itself is under design by an external mechanical/casing development partner (legal identity now confirmed as PT Galaksi Megatama Indonesia — see Section 1.4) and its bill of materials has not yet been provided to the authors.

2.6.c • Material Specification Sheets / Datasheets (Non-Metallic Materials)

Datasheets or supplier conformity declarations for non-metallic materials (enclosure components, seals, insulators, potting compounds) — covering heat/cold resistance, anti-aging, anti-static, flame retardancy, CTI value, and chemical resistance — have not yet been collected.

Per IEC 60079-0:2017, once GLD's specific non-metallic components are identified the applicable requirements are: a Comparative Tracking Index (CTI) class per Table 5/6 appropriate to the equipment's creepage/clearance design; a glow-wire or needle-flame flame-retardancy result per §26.4–26.5; a glass-transition temperature (T_g) safely outside the confirmed -20°C to $+60^{\circ}\text{C}$ ambient operating range (Section 2.3.c); and UV/weathering-resistance data for any component exposed outdoors. These thresholds are cited from the standard itself and have not yet been matched against actual component datasheets.

Non-metallic components in the design that this item will need to cover: the cover-to-body sealing gasket providing the IP66 rating (Section 2.3.c — identified as a rubber gasket in the casing partner's assembly drawing, specific compound/grade not yet specified); the integral seal of the candidate M20×1.5 cable gland — per project direction, specified together with the gland itself from the BP18-1Z reference (Section 2.3.c), rather than as a separately sourced component; and the antenna/SMA feedthrough insulator (material not yet specified). The protective mesh cover over the gas-sensing element is now identified as a stainless-steel wire mesh in the same assembly drawing — being metallic, it falls outside this specific non-metallic-materials checklist item.

Status: Not yet available. Dependent on the enclosure design and material selection, which sits with the external casing development partner (PT Galaksi Megatama Indonesia, Section 1.4) rather than the authors; this data has been formally requested from that partner. Partial candidate data: the BP18-1Z reference drawing (Section 2.6.a) specifies a powder-coating surface treatment with two electrostatic-safety parameters directly relevant to this checklist item — maximum surface charge transfer $<10\text{ nC}$ and maximum surface capacitance $<5\text{ pF}$ (coating thickness $\leq 0.2\text{ mm}$). These are the type of parameter IEC 60079-0 requires for coated/non-metallic surfaces on Ex equipment, but they describe a supplier's candidate component, not a confirmed datasheet for GLD's own enclosure finish.

2.6.d • Manufacturing Process Description

A description of manufacturing processes relevant to explosion-protection safety (enclosure machining accuracy control, explosion-proof surface treatment, welding, potting, die-casting, bonding) has not yet been documented.

Status: Not yet available. This describes the casing partner's (PT Galaksi Megatama Indonesia) manufacturing process, not an internal electronics process — it will need to be obtained from that partner once their process is finalized.

2.6.e • Explosion-Protection Calculations and Explanations (if applicable)

Flame-path (joint gap, length) and free-internal-volume calculations specific to Ex d (Section 2.5, team recommendation) have not yet been performed, pending final enclosure geometry from the casing development partner and a confirmed protection concept with the ExCB.

Status: Not yet available — pending ExCB confirmation of the Ex d protection concept and final enclosure design.

2.6.f • Temperature Group Calculation (Hottest-Point Temperature Estimation)

A preliminary internal surface-temperature measurement has been carried out on a GLD V2 unit using thermocouples at eight locations (plus an ambient reference), under three conditions: normal operation, maximum load, and fan stalled (an expected malfunction of the internal DC cooling fan). Each condition was logged for 180 minutes at an ambient temperature of approximately 24–25°C.

CHANNEL	LOCATION	RATIONALE
T amb	Air around the unit	Reference temperature
TC-1	Hottest MQ sensor body	The sensor heater is the main potential heat source
TC-2	DC fan motor/body	Normal and stalled-fan heating
TC-3	DC/DC converter	Power component — not yet measured
TC-4	Power inductor	Potential hot spot — not yet measured
TC-5	MOSFET / power diode	Potential local heating — not yet measured
TC-6	Hottest PCB area	PCB temperature verification
TC-7	External enclosure surface near the heat source	Candidate external maximum
TC-8	Other external enclosure surface	Confirms the external maximum is not missed

Maximum values recorded over the 180-minute runs, and the same values extrapolated to the +60°C maximum rated ambient (Section 2.5) by adding the measured temperature rise above ambient:

CONDITION	MEAN T AMB	MQ SENSOR BODY (TC-1)	PCB (TC-6)	FAN (TC-2)	EXTERNAL ENCLOSURE (TC-7/8)
Normal operation	24.9°C	44.2 → ≈79°C	52.4 → ≈88°C	36.3°C	32.1 → ≈67°C
Maximum load	24.2°C	55.0 → ≈91°C	58.3 → ≈94°C	30.2°C	34.8 → ≈71°C

CONDITION	MEAN T AMB	MQ SENSOR BODY (TC-1)	PCB (TC-6)	FAN (TC-2)	EXTERNAL ENCLOSURE (TC- 7/8)
Fan stalled (malfunction)	24.8°C	78.4 → ≈114°C	63.4 → ≈99°C	36.3°C	43.9 → ≈79°C

Preliminary reading against T4. The T4 limit is 135°C; with the 5 K margin applied for temperature classes T3–T6 under IEC 60079-0, the effective limit is 130°C. The highest extrapolated value, ≈114°C on the MQ sensor body in the fan-stalled condition, leaves a margin of about 16 K. The heater element itself (in the approximate 200–400°C range typical of this sensor class) sits inside the sensor can and was not measured directly; the measured sensor-body surface is 44–78°C. Under the recommended flameproof concept (Ex d, Section 2.5), the temperature class is governed by the external surfaces of the flameproof enclosure, whose extrapolated maximum is ≈79°C.

Open items before this can be treated as a temperature-class verification:

- TC-3, TC-4 and TC-5 (DC/DC converter, power inductor, MOSFET/power diode) have not been measured; no independent hot-spot location check has been recorded.
- The alarm/transmit condition has not been measured.
- Thermal equilibrium was not demonstrated in every run: the PCB reading under maximum load was still rising in the final hour (46.4 → 55.8 → 58.3°C), above the 2 K/h stability criterion.
- Two short excursions on TC-7 (43.4 and 43.9°C) in the fan-stalled run, with neighbouring readings near 30°C, should be checked for thermocouple contact before the external maximum is finalised.
- Test metadata is not yet recorded: date, operator, unit identification, thermocouple type, supply voltage (the most unfavourable voltage within the rated range should be used), and whether the enclosure was fully closed.
- Extrapolation to +60°C assumes a constant temperature rise above ambient; confirmation requires testing at maximum ambient as part of ExCB type testing.

Status: Partially available — preliminary internal measurement. Indicates T4 is achievable with margin; not a substitute for type testing, and the open items above must be closed first.

2.6.g • Usage and Installation Instructions (Draft)

SUB-ITEM	STATUS	REMARKS
a) Safety warnings	Draft available	Drafted below (18 Sep 2026); not yet reviewed against a selected explosion-protection concept.
b) Installation requirements (cable entry, torque, grounding, cleaning)	Draft available	Procedural text drafted below; the mechanical mounting method itself (U-bolt mounting plate, no drilling/welding — Section 2.3.c) and the specific cable gland/torque values are still pending final component selection.

SUB-ITEM	STATUS	REMARKS
c) Operating instructions and maintenance requirements	Draft available	Drafted below, alongside a firmware command/operation reference (Serial, MQTT, LoRa protocol) used for engineering commissioning. Ex-specific maintenance frequency/content is not yet defined pending protection-concept selection.

Status: Draft text available (source: internal working draft, Rev. 18 Sep 2026). This is a draft for review, not a finalized, Ex-approved installation and operation manual. It has not been checked against a selected type of protection (Ex d/e/i, Section 2.6.e), a confirmed temperature class (Section 2.6.f), or finalized cable-entry/enclosure hardware. It also feeds the standalone Instruction_Manual_GLD deliverable.

2.6.g.a • Safety Warnings (Draft)

- Only trained and authorized personnel may install, commission, operate, inspect, or maintain the detector.
- Isolate and verify the 24 VDC supply before opening the enclosure or changing wiring. Apply the site lockout/tagout procedure.
- Use only the approved production power arrangement: 24 VDC.
- Perform alarm tests with appropriate site controls and hearing protection where audible devices are connected.
- Perform sensor nulling only in confirmed clean air. Accept the operation only when all eight sensors report a complete 8/8 result; otherwise correct the cause and repeat.
- Do not drill, weld, enlarge cable entries, bypass protective devices, or modify the enclosure without written engineering and Ex approval.
- Remove the unit from service if there is visible damage, water ingress, an unresolved sensor fault, failed alarm proof, or a loss of protective-earth continuity.

2.6.g.b • Detailed Installation Requirements (Draft)

Cable entry method: Use only cable glands, adaptors, blanking elements, and seals approved for the final protection concept, certificate, thread form, IP rating, cable outer diameter, and installation temperature. Do not create new entries, enlarge an entry, leave unused entries open, or substitute a general-purpose gland. Maintain the manufacturer sealing and clamping sequence. Record the installed component part number and certificate reference.

Torque requirements: Torque values shall comply with the approved enclosure, terminal, cable-gland, and blanking-plug manufacturer instructions. (Specific torque figures are not yet available — dependent on final hardware selection.)

Grounding requirements: Connect the enclosure to the approved protective-earth system using the designated grounding terminal. Verify continuity before energization.

Cleaning requirements: For routine cleaning, isolate the equipment as required by site procedure and use a soft lint-free cloth lightly dampened with clean water or an approved mild cleaner. Do not use abrasives, aggressive solvents, dry compressed air, high-pressure spray, or tools that can damage the enclosure, gasket, label, antenna, or cable entries. Keep liquid out of all openings.

2.6.g.c • Operating Instructions and Maintenance Requirements (Draft)

Before each operating period, check power, enclosure condition, diagnostic status, communication status, and the absence of active faults. Treat an alarm as a real process-safety event until the site response procedure establishes otherwise. If the detector reports a fault or loses required communication, notify the responsible operator and place

the affected detection point in the site impairment process. Do not change firmware, model files, alarm thresholds, or calibration parameters from the field without controlled approval and a documented rollback path.

Three operating modes are distinguished:

MODE	PURPOSE	USE RESTRICTION
Inference	Normal gas-class inference and alarm operation	Production operating mode after commissioning
Dataset	Engineering / data-collection activity	Not a substitute for approved detection operation
Nulling	Sensor baseline / calibration activity	Clean air only; perform under approved procedure; require complete 8/8 result

2.6.h • Nameplate Information

Nameplate artwork cannot yet be finalized: it depends on the certificate number, protection marking, temperature class, ambient range, IP rating, and serialization scheme — none of which has been assigned yet.

Status: Not yet available.

2.6.i • Ex Component Certificates

No components in this design currently hold an Ex component certificate. As noted in Section 2.3.b, the processing unit (ESP32-S3-WROOM-1U-N16R8) and the communication module (E22-900MM22S) hold RF/EMC certifications (FCC, TELEC, CE, RoHS) — these are not Ex component certificates and do not satisfy this item.

Status: Not yet available.

3. Sample Information

Model, serial number, and sample status (whether the unit can be powered on and operated), together with any required test fixtures or auxiliary equipment.

3.1 • Model, Serial Number, and Status

No formal sample register or dossier for ExCB submission has been established. Internal engineering and bench testing reference the “GLD V2” board configuration under firmware environment gld_v2; a discrete unit serial-numbering scheme for certification samples has not yet been implemented.

Status: Not yet available. The product is still being built from scratch at prototype/development stage; no finalized, serialized unit yet exists to register as a submission sample. This item becomes actionable once a build reaches a stable, submission-ready configuration.

3.2 • Test Fixtures and Auxiliary Equipment

A draft internal functional test plan identifies the minimum equipment anticipated for bench-level verification (this plan has not yet been executed and is not evidence of lab readiness):

TEST GROUP	MINIMUM EQUIPMENT
Firmware / serial	Engineering commissioning tool or serial terminal; firmware package with recorded version
I2C / ADC / DAC	No additional equipment for protocol-level acknowledgement/readback; multimeter or oscilloscope where physical voltage must be substantiated
Power / watchdog timer	Controlled 24 V supply, an applicable battery source, multimeter, and oscilloscope/logic analyzer
Alarm	The actual alarm load, multimeter/oscilloscope, and hearing protection if an audible buzzer is fitted
LoRa	At least one counterpart Cluster Head/Gateway device with recorded configuration
RS-485 / Modbus	An RS-485/USB-RS485 master with proper termination and the agreed register map

Status: Draft plan only — not yet executed, and not evidence of ExCB/laboratory test readiness.

A second, commissioning-oriented equipment list is drafted separately (source: internal working draft, Rev. 18 Sep 2026), organized by physical test apparatus rather than by test group:

EQUIPMENT	PURPOSE
Regulated 24 VDC power supply	Provides a stable 24 VDC supply to the GLD during commissioning and functional testing.
Digital multimeter	Measures the actual voltage, verifies polarity and continuity, and supports grounding/bonding verification.
Calibrated gas source / gas test kit	Provides the required gas types to verify that the sensors respond correctly when exposed to the target gases.
Gas test chamber	Provides a controlled gas-exposure environment so that gas concentration, exposure time, and test conditions can be reproduced.
Alarm load / relay simulator	Physically verifies that the alarm output and connected electrical load operate correctly.
LoRa peer / gateway	Verifies that data transmitted by the GLD is successfully received by the intended system.
Laptop and approved software	Used to review diagnostic information, configuration, logging, firmware identification, and to store commissioning and test evidence.
Calibrated torque screwdriver	Ensures that glands, terminals, enclosure fasteners, and grounding connections are tightened to the approved torque values.

This is a working document, prepared in stages, drafted in direct reference to the IECEx/ATEX Certification Information Requirements (original English/Mandarin version issued by the certification body). Data sources: the official product technical datasheet (Revision 4.0), internal technical specification documentation, EMC parameter measurement data, and product photography. Fields marked “Pending confirmation” are not yet final and must not be relied upon for procurement or certification purposes without further verification.